Universitatea Națională de Știință și Tehnologie Politehnica
București
Facultatea de Electronică, Telecomunicații și Tehnologia
Informației

Developing a System for Physiological Signal Acquisition and Analysis for Sleep Quality Assessment

Diploma Project

presented as a partial requirement for obtaining the title of
Engineer in the field of *Electronics, Telecommunications and Information Technology*

Scientific coordinator
Ph. D. Ovidiu GRIGORE

Graduate
Duta Alexandru-Vlad

2026

National University of Science and Technology POLITEHNICA Bucharest
Faculty of Electronics, Telecommunications and Information Technology
Study program **ELAen**

Anexa 1

DIPLOMA THESIS
of student **DUȚĂ C.C. Alexandru-Vlad, 442F**

1. Thesis title: Dezvoltarea unui sistem pentru achiziția și analiza semnalelor fiziologice în vederea evaluării calității somnului

2. The student's original contribution(not including the documentation part):
Scopul lucrării este de a proiecta și implementa un sistem de achiziție, prelucrare și analiză a semnalelor fiziologice în vederea evaluării calității somnului. Sistemul este compus dintr-o parte hardware bazata pe platforma Raspberry Pi, care colectează date de la senzori, precum PPG, accelerometru, senzor de temperatură, microfon, banda toracică. Partea software are ca scop prelucrarea și clasificarea semnalelor obținute în scopul determinării fazelor de somn și a calculării unui scor global al calității somnului.

3. Academic courses the thesis is based on:
Lucrarea trebuie să conțină următoarele etape:
1. Studiul cu privire la tipologia senzorilor utilizați (PPG, accelerometru, senzor de temperatura, microfon low-power, senzor toracic) și analiza parametrilor fiziologici derivați din aceștia.
2. Proiectarea și implementarea sistemului hardware de achiziție, bazat pe platforma Raspberry Pi și interconectarea senzorilor selectați.
3. Realizarea unei baze de date prin colectarea și stocarea datelor/semnalelor achiziționate în timpul somnului, precum și rezultatele unui chestionar completat dimineață de subiect conținând opinii subiectiv privind calitatea somnului ce vor fi folosite pentru validarea rezultatelor.
4. Implementarea unor algoritmi pentru prelucrare, filtrare și extragerea parametrilor relevanți din semnalele fiziologice, precum și clasificarea datelor utilizând algoritmii K-NN si K-means .
5. Analiza și compararea rezultatelor obținute, evaluarea performanțelor celor două metode de clasificare și calcularea scorului global al calității somnului pe baza fazelor detectate și a evenimentelor respiratorii (apnee).
6. Concluzii cu privire la date experimentale, prelucrarea acestora, observații privind utilizarea și performanțele algoritmilor, posibilități de îmbunătățiri.

4. Resources and bibliography:

5. Thesis registration date: 2025-12-12 13:48:56

Thesis advisor(s),
Prof. dr. ing. Ovidiu GRIGORE

Student,
DUȚĂ C.C. Alexandru-Vlad

Departament director,
Ș.L. dr. ing Bogdan FLOREA

Dean,
Prof. dr. ing. Mihnea UDREA

Validation code: **bfc6f32dd6**

Anexa 5

Declarație de onestitate academică

Prin prezenta declar că lucrarea cu titlul “ *Dezvoltarea unui sistem pentru achiziția și analiza semnalelor fiziologice în vederea evaluării calității somnului (Developing a System for Physiological Signal Acquisition and Analysis for Sleep Quality Assessment)*”, prezentată în cadrul Facultății de Electronică, Telecomunicații și Tehnologia Informației a Universității “Politehnica” din București ca cerință parțială pentru obținerea titlului de *Inginer* în domeniul *Electronică și Telecomunicații*, programul de studii *Electronică aplicată (în limba engleză)* este scrisă de mine și nu a mai fost prezentată niciodată la o facultate sau instituție de învățămînt superior din țară sau străinătate.

Declar că toate sursele utilizate, inclusiv cele de pe Internet, sunt indicate în lucrare, ca referințe bibliografice. Fragmentele de text din alte surse, reproduse exact, chiar și în traducere proprie din altă limbă, sunt scrise între ghilimele și fac referință la sursă. Reformularea în cuvinte proprii a textelor scrise de către alți autori face referință la sursă. Înțeleg că plagiatul constituie infracțiune și se sancționează conform legilor în vigoare.

Declar că toate rezultatele simulărilor, experimentelor și măsurătorilor pe care le prezint ca fiind făcute de mine, precum și metodele prin care au fost obținute, sunt reale și provin din respectivele simulări, experimente și măsurători. Înțeleg că falsificarea datelor și rezultatelor constituie fraudă și se sancționează conform regulamentelor în vigoare.

București, *23.06.2026*

Absolvent, *Duta Alexandru-Vlad*

(semnătura în original)

Contents

List of Figures

List of Tables

List of Acronyms

AC – Alternating Current
ADC – Analog-to-Digital Converter
BPM – Beats Per Minute
DC – Direct Current
ECG – Electrocardiogram
EEG – Electroencephalogram
EMG – Electromyogram
EOG – Electrooculogram
ESP – Espressif Systems Platform
FFT – Fast Fourier Transform
FIFO – First In, First Out
GPIO – General Purpose Input/Output
HF – High Frequency
HRV – Heart Rate Variability
HTTP – Hypertext Transfer Protocol
I2C – Inter-Integrated Circuit
IMU – Inertial Measurement Unit
IR – Infrared
JSON – JavaScript Object Notation
KNN – K-Nearest Neighbors
LF – Low Frequency
LiPo – Lithium Polymer
MEMS – Micro-Electro-Mechanical Systems
NREM – Non-Rapid Eye Movement
OSA – Obstructive Sleep Apnea
PCA – Principal Component Analysis
PPG – Photoplethysmography
PSG – Polysomnography
REM – Rapid Eye Movement
RMSSD – Root Mean Square of Successive Differences
SCL – Serial Clock Line
SDA – Serial Data Line
SNR – Signal-to-Noise Ratio
SpO2 – Peripheral Oxygen Saturation
SQLite – Self-Contained SQL Database Engine
WAL – Write-Ahead Logging
WHO – World Health Organization
WiFi – Wireless Fidelity

Introduction

Sleeping is a normal body process that allows your body and brain to rest, occupying roughly a third of your life. During sleep, some key things happen: energy conservation and storage, meaning that your body uses less energy, and self-repair and recovery, which fall into the deep sleep category, known as NREM. On the other hand, REM sleep helps with brain maintenance, your brain reorganizes memories and learned information, and processes emotional experience. When these stages are affected, the consequences can appear in many forms, such as fatigue, poor concentration, mood changes, reduced cognitive performance, and a higher risk of long-term health problems.
Sleep disorders are widespread. The WHO estimates that between 10% and 30% of adults globally have chronic insomnia. Obstructive sleep apnea , repeated airway obstruction during sleep, affects hundreds of millions and remains largely undiagnosed because the standard diagnostic procedure is both expensive and hard to access. Beyond people diagnosed with a certain sleep disorder, there are many cases where poor sleep accumulates, and it is often ignored until it starts to influence daily activity and performance.

Polysomnography, or PSG, is the most reliable method used today for sleep evaluations. It consists of an overnight examination that simultaneously records brain activity, eye movement, muscle activity, heart activity, breathing effort, and blood oxygen saturation. Scientific literature classifies sleep into stages such as N1, N2, N3, and REM, usually by analyzing 30-second intervals based on these overnight evaluations. PSG must be performed in a specialized laboratory, requires multiple sensors and electrodes attached to the patient, can be uncomfortable, and is usually expensive. Because of these limitations, and besides the fact that it offers accurate and clinically validated results, it is not a very practical solution for monitoring sleep.

The Apple Watch, Fitbit, Garmin watches, and the Oura Ring are some examples of wearable devices that can collect physiological data during sleep outside the laboratory, using small, low-power sensors attached to the wrist or fingers. These devices have gained momentum in recent years and made sleep tracking more accessible and easier to use. However, commercial solutions are not transparent about the algorithms they use and also do not provide access to raw sensor data. Also, studies have shown that wearable devices have limited accuracy for distinguishing deep sleep from REM sleep, these two being the most important stages for physical and cognitive recovery. Cost can also be a relevant factor, as these devices are not very budget-friendly for low-cost research.

This project aims to address these limitations by presenting the implementation, design, and validation of an open-source wearable system for sleep tracking. The system is built around an ESP32-C3 microcontroller placed on the wrist, together with five physiological sensors. The main sensor is the MAX30102, used for photoplethysmography and pulse oximetry, alongside the MPU6050, which measures movement and rotation, the BMP280, used for skin temperature estimation, a flex sensor connected through an ADS1015 analog-to-digital converter to monitor breathing, and the MAX4466 microphone, which is used to detect snoring. A Raspberry Pi server collects the raw data from the ESP32 through WiFi, stores it in a SQLite database, and makes it available for the analysis pipeline. The system is designed to be easy to modify and extend, while also giving full access to raw data.

The software pipeline analyzes the database and extracts 21 physiological features from 30-second windows. Using machine learning methods, these features are used to analyze sleep patterns. K-Means clustering is applied to identify possible sleep stage groups without using predefined labels, while K-Nearest Neighbors is used to estimate sleep stages in a probabilistic way. Based on this analysis, heart rate variability, respiratory rate, and motion are the most useful features for differentiating sleep stages. The system also calculates a sleep quality score based on six physiological components. This score is compared with morning questionnaire answers from the monitored subject, in order to relate the measured data to the person's own perception of sleep quality.

The thesis is organized into six chapters. Chapter 1 presents the basic concepts of sleep physiology, the main characteristics of each sleep stage, and a review of existing sleep monitoring solutions. Chapter 2 describes the hardware structure of the system, including the role and integration of each sensor. Chapter 3 covers the data acquisition software, both on the ESP32 side and on the Raspberry Pi server. Chapter 4 explains how the raw signals are preprocessed and how the physiological features are extracted. Chapter 5 presents the machine learning methods used for sleep stage analysis, along with the sleep quality score calculation. Chapter 6 discusses the experimental results obtained from overnight recordings, evaluates the system's performance, and compares the objective measurements with the subjects' own perception of sleep quality.

Chapter 1. Theoretical Background

This chapter covers the theoretical background needed to understand the rest of the thesis. It starts with sleep physiology and what happens during each stage, then moves to the physiological signals relevant for sleep monitoring and ends with a review of existing solutions and their main limitations.

1.1 Sleep Physiology and Sleep Stages

Sleep is a biologically active process organized into repeating cycles of distinct stages, each with its own characteristic brain activity, muscle tone, eye movements, and autonomic behavior. The American Academy of Sleep Medicine classifies sleep into four stages: three NREM stages (N1, N2, N3) and REM [3]. A typical adult night contains four to six cycles of roughly 90 minutes each, with N3 concentrated in the first half of the night and progressively longer REM periods toward morning [4].

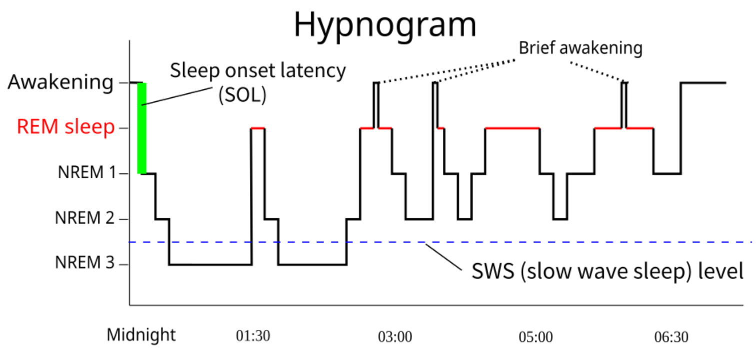


Figure 1.1 , Typical adult hypnogram showing four complete sleep cycles over an 8-hour night. N3 sleep predominates in the first two cycles while REM periods lengthen progressively toward morning. Adapted from [3].

1.1.1 NREM Sleep, Stages N1, N2, N3

N1 is the transition between wakefulness and sleep, accounting for about 5-10% of total sleep time [3]. The alpha rhythm of relaxed wakefulness is replaced by lower-amplitude theta activity, muscle tone drops, and the person wakes easily. N1 has limited restorative function on its own.

N2 accounts for the largest fraction of sleep, typically 45-55% in healthy adults [4]. It is defined by two EEG features: sleep spindles (12-15 Hz bursts lasting 0.5-3 seconds) and K-complexes (high-amplitude biphasic waveforms that suppress arousal). Spindles are specifically linked to memory consolidation [1]. Heart rate and body temperature both begin to fall during this stage.

N3, or slow-wave sleep, is the deepest stage. Delta waves (0.5-4 Hz) dominate, and young adults typically spend 20-25% of the night here, though this declines with age [4]. Heart rate, respiratory rate, and blood pressure are at their lowest. Growth hormone secretion peaks and immune activity is highest during N3 [5]. After sleep deprivation, the body prioritizes N3 recovery the following night.

1.1.2 REM Sleep

REM is neurologically the most active sleep stage. The EEG resembles wakefulness in pattern, but skeletal muscle tone is actively suppressed by brainstem inhibition [6], which explains why most dreaming happens here without physical enactment. REM accounts for

20-25% of total sleep time, with episodes growing from a few minutes in the first cycle to 30-45 minutes by morning.

The stage plays an important role in emotional memory processing and in consolidating procedural and associative learning [1]. Autonomic behavior is irregular during REM, heart rate and breathing fluctuate, intermittent sympathetic bursts occur, and thermoregulation is temporarily suspended [22].

1.1.3 Sleep Cycles and Architecture

Sleep architecture describes the composition, timing, and sequencing of sleep stages across a full night, usually represented as hypnogram. Normal adult sleep follows a well-characterized pattern [3].

Sleep onset takes 10-20 minutes for healthy individuals. The first N3 period starts around 30-45 minutes after sleep onset and can last 20-40 minutes in young adults, followed by a brief first REM episode of only 5-10 minutes. In later cycles, N3 gets shorter and is replaced by N2, while REM periods grow progressively longer [4].

Brief awakenings under 2-3 minutes are normal and occur 5-15 times per night, more often with age, sleep disorders, or noise. The most consistent age-related change is a reduction in N3, along with increased fragmentation and lower sleep efficiency. These variations are relevant when interpreting monitoring results and when designing the sleep quality scoring algorithm described in Chapter 5.

Table 1.1 summarizes typical sleep parameters for healthy young adults and the thresholds associated with clinical concern, based on AASM guidelines [3].

Table 1.1 , Normal adult sleep architecture parameters and clinical concern thresholds. Adapted from AASM guidelines [3].

Parameter	Typical Value (Young Adults)	Clinical Concern If
Total Sleep Time	7–9 hours	< 6 hours chronically
Sleep Efficiency	> 85%	< 75%
Sleep Onset Latency	< 20 minutes	> 30 minutes
N1 (%)	5–10%	> 15% (fragmented sleep)
N2 (%)	45–55%	< 30% or > 65%
N3 (%)	20–25%	< 10% (poor recovery)
REM (%)	20–25%	< 15% (memory impairment risk)
Wake After Sleep Onset	< 30 minutes	> 45 minutes
Number of cycles	4–6	< 3 (short sleep)

1.2 Physiological Signals During Sleep

Without EEG, sleep stage classification depends on peripheral signals that indirectly reflect what the brain and autonomic nervous system are doing. For a wrist-worn device used at home, those signals also have to be non-invasive, continuous, and small enough to collect without disturbing sleep. The six signals used in this thesis are described below, covering how each one is generated physiologically and why it carries information about sleep stage.

1.2.1 Photoplethysmography (PPG) and Heart Rate

Photoplethysmography detects blood volume changes in peripheral tissue by shining light onto the skin and measuring how much is reflected back [17]. Pulse oximetry uses two wavelengths, red and infrared, because oxyhemoglobin and deoxyhemoglobin absorb them differently. Each heartbeat sends a pressure wave through the arteries that briefly increases blood volume at the sensor site, producing a periodic waveform synchronized with the cardiac cycle.

The signal has an AC component that pulses with each beat and a DC component reflecting average tissue absorption. Heart rate comes from peak detection on the AC part, while the ratio of AC to DC amplitudes at the two wavelengths gives SpO2 via the Beer-Lambert law [18]. Across sleep stages, heart rate decreases from wakefulness through N1 and N2, reaches its minimum during N3, and becomes irregular during REM with intermittent autonomic surges.

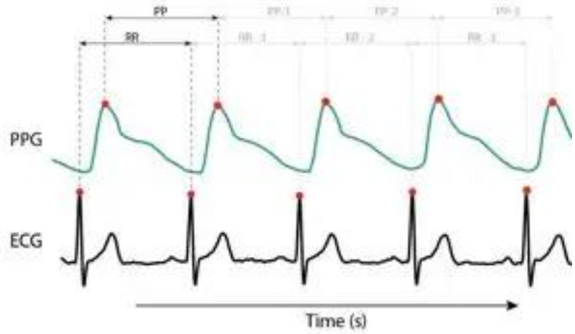


Figure 1.2 , Anatomy of a reflectance PPG waveform acquired at the wrist. The AC component (pulsatile variation) is superimposed on a large DC baseline. SP: systolic peak; DN: dicrotic notch; DP: diastolic peak. Adapted from Allen (2007) [17].

PPG peaks are delayed relative to ECG R-peaks by the pulse transit time; PP intervals extracted from PPG are used as surrogates for RR intervals in HRV analysis.

1.2.2 Heart Rate Variability (HRV)

Heart rate variability is the beat-to-beat variation in the intervals between successive heartbeats, measured either between R-peaks in an ECG or between systolic peaks in a PPG signal [19]. High HRV indicates strong parasympathetic activity and good cardiac responsiveness; low HRV is associated with autonomic dysfunction and cardiovascular risk.

HRV is analyzed in the time domain, frequency domain, or using non-linear methods. The most commonly used time-domain metric is RMSSD, the root mean square of successive RR interval differences, a specific marker of parasympathetic activity that has been extensively validated [19]. In the frequency domain, the LF band (0.04-0.15 Hz) reflects mixed sympathetic and parasympathetic activity, the HF band (0.15-0.40 Hz) corresponds to respiratory sinus arrhythmia, and the LF/HF ratio tracks sympathovagal balance.

During sleep, HRV follows a pattern closely tied to sleep stage. Parasympathetic dominance during NREM produces elevated RMSSD and high HF power, with the strongest vagal activity during N3 [20]. REM reverses this: intermittent sympathetic surges reduce HRV and push the LF/HF ratio up. This NREM-REM contrast is one of the most consistent physiological distinctions available to a wearable system and drives a significant part of the classification performance described in Chapter 5.

1.2.3 Motion and Accelerometry

Body movement during sleep is a direct indicator of sleep stage. Wrist actigraphy has been used clinically for decades as a low-cost alternative to PSG for estimating sleep-wake cycles [13]. Wakefulness produces the highest motion activity, while both NREM and REM show greatly reduced movement. REM atonia typically gives the lowest absolute levels, though brief involuntary muscle twitches during phasic REM create transient spikes [13].

For sleep monitoring purposes, combining translational acceleration with rotational velocity provides complementary information: gross positional shifts appear primarily in the accelerometer axes, while subtler torso rotations during sleep-stage transitions are more visible in the gyroscope channels.

1.2.4 Respiratory Rate

Respiratory rate changes in a characteristic way across sleep stages. In healthy adults, the typical range during wakefulness is 12-20 breaths per minute, slowing to approximately 10-14 during N3 [3]. REM breathing is irregular, reflecting the desynchronized neural activity of that stage.

In wearable devices, respiratory rate is measured through indirect methods such as mechanical movement sensing or impedance pneumography [21]. The respiratory signal also reveals apnea events, defined as breathing pauses of 10 seconds or more. Obstructive Sleep Apnea affects an estimated 9-38% of the adult population, with higher rates in men and overweight individuals [24]. Repeated apnea events fragment sleep, suppress N3, and carry significant cardiovascular consequences [26]. Detection of respiratory pauses is part of both the feature extraction pipeline and the sleep quality score described in Chapter 5.

1.2.5 Skin Temperature

In the hours before sleep onset, peripheral skin temperature rises as extremity blood vessels dilate, helping dissipate core heat, a process necessary for sleep initiation [22]. Core temperature reaches its minimum in the early morning and begins rising before waking. N3 shows the greatest peripheral vasodilation, while REM partially reverses this pattern because thermoregulation is suspended during phasic events [22]. Temperature alone has limited stage-discriminative power, but it adds useful information for detecting sleep onset and wakefulness, particularly when combined with other signals.

1.2.6 Acoustic Signals and Snoring

Snoring comes from vibration of soft tissue in the upper airway when reduced muscle tone during sleep allows partial airway collapse and increases airflow turbulence [24]. It is also the main symptom of obstructive sleep apnea (OSA), where complete or near-complete collapse produces repeated apnea episodes, defined as cessation of airflow for 10 seconds or more, associated with oxygen desaturation and arousal. OSA carries increased risk of hypertension, cardiovascular disease, and stroke [26].

Automated snoring detection from microphone recordings has been explored as a low-cost screening approach for OSA [25]. In the context of a wrist-worn monitoring system, acoustic detection complements the respiratory and SpO2 signals by providing a direct indicator of upper-airway obstruction events, which would otherwise go undetected from peripheral measurements alone.

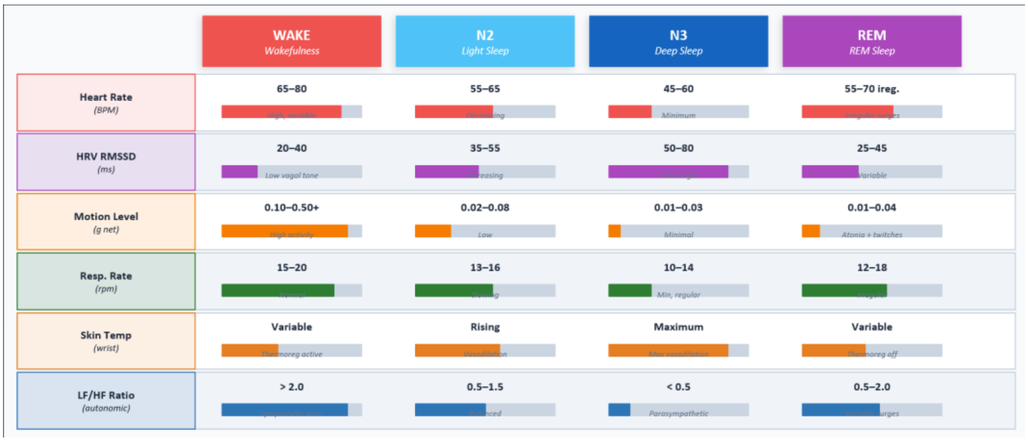


Figure 1.3 , Representative physiological signal traces during four sleep stages. Note the progressively lower heart rate and motion from Wake to N3, the respiratory irregularity during REM, and the increased HRV during N3.

Table 1.2 , Typical physiological signal ranges across sleep stages. Values represent approximate ranges for healthy adults.

Signal	Wake	N2 (Light)	N3 (Deep)	REM
Heart Rate (BPM)	65–80	55–65	45–60	55–70 (irregular)
HRV RMSSD (ms)	20–40	35–55	50–80	25–45 (variable)
Motion Level (g net)	0.10–0.50+	0.02–0.08	0.01–0.03	0.01–0.04
Respiratory Rate (rpm)	15–20	13–16	10–14	12–18 (irregular)
Skin Temp (°C, wrist)	Variable	Stable rising	Maximum	Variable
Snoring (relative)	Absent	Possible	Possible	Possible

1.3 Existing Sleep Monitoring Systems

The field of sleep monitoring has undergone a dramatic transformation over the past three decades, progressing from fully laboratory-based clinical examinations requiring overnight hospitalization toward increasingly miniaturized and affordable solutions capable of operating continuously in the home environment. Understanding the technical capabilities, clinical validity, and practical limitations of existing approaches provides essential context for evaluating the contributions of the system developed in this thesis.

1.3.1 Clinical Polysomnography (PSG)

PSG is the reference standard for sleep evaluation [3]. A full recording includes at minimum six EEG channels, two EOG channels, chin and leg EMG, ECG, nasal airflow, respiratory effort belts, and pulse oximetry. Setup takes 1-2 hours by a trained technician, and manual epoch-by-epoch scoring takes another 1-2 hours after the recording ends.

The practical limitations are well known. PSG requires a dedicated laboratory and is expensive. The lab environment itself alters the sleep being measured: the first-night effect, a systematic change in sleep architecture caused by unfamiliar surroundings, has been documented since the 1960s and reduces the validity of single-night recordings [9].

Inter-rater agreement is generally good for NREM-REM discrimination but drops for N1-N2 transitions. These constraints make PSG impractical for routine monitoring or longitudinal research.

1.3.2 Commercial Wearables

Consumer devices such as Fitbit, Apple Watch, Garmin, and Oura Ring use combinations of PPG, accelerometry, skin temperature, and HRV to estimate sleep stages. Validation studies show good sensitivity for sleep-wake discrimination, typically above 85%, but performance drops substantially for finer stage differentiation [10]. Distinguishing N3 from N2 or REM from light NREM, which are the most important distinctions clinically, gives epoch-by-epoch agreement with PSG in the range of 60-75%. Without EEG, the signals that actually define sleep stages, slow oscillations, sleep spindles, K-complexes, desynchronized REM activity , are simply not measurable at the wrist.

The second problem is transparency. Classification algorithms are proprietary and unpublished, raw sensor data is inaccessible, and outputs come only as post-processed summaries through manufacturer apps [11]. This makes commercial devices unsuitable for research where reproducibility and access to raw signals matter.

1.3.3 Limitations of Existing Solutions and Motivation

PSG is accurate but laboratory-bound and expensive. Commercial wearables are affordable but use opaque algorithms, block access to raw data, and cannot reliably distinguish N3 from REM. Research-grade portable systems are more transparent but still costly and technically demanding.
The system described in this thesis addresses all three problems. Five sensors, PPG with SpO2, six-axis inertial measurement, skin temperature, thoracic respiratory effort, and acoustic monitoring , are integrated into a single wrist-worn device built from off-the-shelf components costing under 100 euros. The design is fully open-source, raw data is stored in a SQLite database accessible for independent analysis, and the processing pipeline implements and compares two classification approaches: K-Means clustering and K-Nearest Neighbors. Sleep stage estimates are validated against subjective morning questionnaire responses collected with the Pittsburgh Sleep Quality Index [8].

	Clinical PSG *Laboratory Standard*	Commercial Wearables *Fitbit / Apple Watch / Oura*	This Thesis System *ESP32 + Raspberry Pi*
Hardware Cost	✗ €500–2,000/night *Lab + technician fee*	✓ €100–500 *One-time purchase*	★ < €100 *All components combined*
Sleep Stage Accuracy	★ Gold standard *Full EEG, EOG, EMG*	△ 65–75% *Limited N3 vs REM discrim.*	✓ ~70–85% target *Multi-sensor + ML pipeline*
Raw Data Access	★ Full access *All channels available*	✗ None / limited *Proprietary, no raw data*	★ Full access *SQLite DB, open pipeline*
Algorithm Transparency	★ AASM published *Fully published criteria*	✗ Proprietary *Undisclosed, not peer-reviewed*	★ Open-source *Full code available*
Home / Longitudinal Use	✗ Not practical *Lab only, 1 night typical*	★ Excellent *Multi-night, continuous*	★ Excellent *WiFi-based, multi-night*
Number of Sensors	★ 20–25 sensors *EEG, EOG, EMG, ECG, resp, SpO2*	△ 2–3 sensors *PPG + accelerometer + temp*	✓ 5 sensors *PPG, IMU, temp, resp, mic*
Subject Comfort	✗ Poor *Many electrodes, lab environment*	★ Excellent *Watch-like, non-obtrusive*	✓ Good *Wrist-worn bracelet*
Subjective Validation	△ Not included *Clinical diagnosis focus*	✓ App survey *Basic morning questions*	★ Questionnaire *Pearson correlation analysis*

Rating: ✗ Not available / Poor △ Limited / Partial ✓ Good ★ Excellent

Figure 1.4 — This system combines the data transparency of clinical PSG with the home-use convenience and low cost of commercial wearables.

Figure 1.4 , Comparison of sleep monitoring approaches across key dimensions. The proposed system aims to combine the raw data accessibility of clinical PSG with the home-use convenience of commercial wearables at a fraction of the cost of research-grade portables.

Chapter 2. System Architecture and Hardware Design

This chapter presents the hardware architecture of the wearable sleep monitoring system. Five low-cost sensors are attached to an ESP32-C3 microcontroller that collects physiological data during sleep and transmits it to a Raspberry Pi server through WiFi, where it is stored in a SQLite database and processed by the analysis pipeline. The main design goals were low cost, portability, reproducibility, low power consumption, and access to raw sensor data.

2.1 System Overview

The system is built around a wrist-worn bracelet that continuously acquires physiological data during sleep and transmits it wirelessly to a Raspberry Pi server. The ESP32-C3 acts as the central acquisition unit, reading data from five sensors over I2C and one analog input, assembling everything into a combined JSON packet, and sending it to the server via HTTP POST every 2 seconds. The Raspberry Pi receives the data, stores it in a SQLite database, and makes it available for the analysis pipeline. The design was kept modular so each sensor can be individually enabled, reconfigured, or replaced without affecting the rest of the system.
The five sensors cover the physiological signals relevant for sleep monitoring: the MAX30102 for PPG and SpO2, the MPU6050 for motion and rotation, the BMP280 for skin temperature, a flex sensor with an ADS1015 ADC for respiration, and a MAX4466 microphone for snoring detection. Each sensor is described in detail in Section 2.3.

2.1.1 Hardware Architecture Diagram

Figure 2.1 shows the full hardware architecture of the system. The ESP32-C3 sits at the center of the design, connected to all five sensors through a shared I2C bus, with the exception of the MAX4466 microphone which uses an analog input. The sensors are grouped on a common prototyping board alongside the ADS1015 ADC, which handles the analog output of the flex sensor. Power is supplied by a 2500 mAh 3.7 V LiPo battery through a TP4056 charging module, connected to the 3.3 V rail of the ESP32-C3. Data is transmitted wirelessly via WiFi using HTTP POST to the Raspberry Pi server, which runs the data collection server and stores all incoming records in a SQLite database. The wire color convention used in the diagram is: red for VCC, black for GND, yellow for SCL, orange for SDA, and white for the MAX4466 analog output.

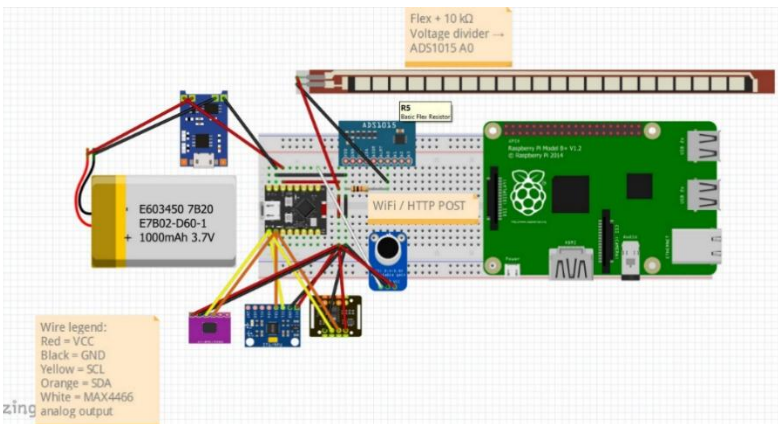


Figure 2.1 , Hardware architecture of the wearable sleep monitoring system.

2.1.2 Data Flow

Each sensor is read at its own rate: MAX30102 at 12.5 Hz effective, MPU6050 at 10 Hz, flex sensor at 4 Hz, BMP280 at 1 Hz, and microphone at 200 Hz in one-second blocks. All samples accumulated since the previous send are packaged into a single combined HTTP POST to the /data endpoint every 2 seconds. Sending all sensor data in a single combined request was necessary because the ESP32-C3 is single-core; the architectural implications of this constraint are described in Section 3.1.3.

On the server side, each incoming record is written to the appropriate SQLite table under a threading lock to prevent write conflicts. Data is stored in raw form without modification, so the sleep pipeline can process it at any point after the session ends.

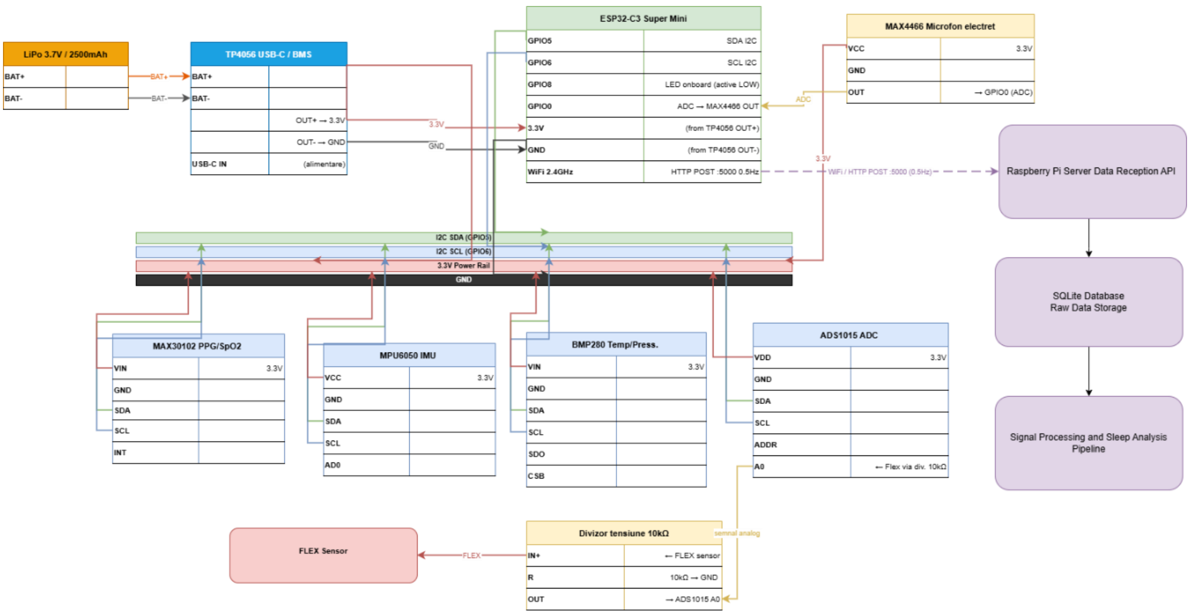


Figure 2.2 , Block diagram of the system architecture, showing the data flow from sensors through the ESP32-C3 acquisition unit to the Raspberry Pi server and the analysis pipeline.

2.2 Microcontroller, ESP32-C3 Super Mini

The ESP32-C3 Super Mini is a compact development board based on Espressif's ESP32-C3 chip. It was chosen for this project for three reasons: it integrates WiFi connectivity and enough GPIO pins for all five sensors in a form factor small enough to fit on the wrist, its RISC-V core at 160 MHz handles sensor reading, JSON assembly, and HTTP transmission without bottlenecks, and it is low-cost and widely available. The C3 variant was preferred over the standard ESP32 for its smaller footprint and lower power consumption, both relevant for overnight battery-powered operation [32].

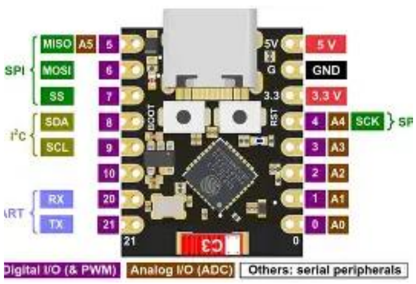


Figure 2.3 , ESP32-C3 Super Mini development board.

2.2.1 WiFi Communication and HTTP Protocol

After boot, the ESP32-C3 connects to the local WiFi network and synchronizes its internal clock via NTP before starting data acquisition. This step is necessary because the board has no real-time hardware clock and without NTP, timestamps in the database would be meaningless. Once connected, the microcontroller waits for the Raspberry Pi server to become available by polling the /status endpoint and only starts sending data after receiving confirmation that the server is active.

All data is sent in a single HTTP POST to the /data endpoint every 2 seconds. The JSON packet includes the latest sensor readings, the PPG samples accumulated since the previous send, the flex buffer collected at 4 Hz, and the microphone block if one is ready. Keeping everything in one request was necessary because the ESP32-C3 is single-core and each HTTP POST blocks the main loop for approximately 500 ms, so sending separate requests for each data type would cause significant sample loss in the PPG, flex, and microphone buffers.

2.2.2 Power Supply, TP4056 + LiPo Battery

The system is powered by a 2500 mAh 3.7 V lithium polymer battery charged through a TP4056 module. The TP4056 handles both charging management and overcurrent protection. Its output is connected to the 3.3 V pin of the ESP32-C3 rather than the 5 V pin, which avoids passing current through the board's internal regulator and reduces unnecessary power dissipation. One important constraint is that the battery and the USB cable must not be connected at the same time, since both feed the same voltage rail and connecting both simultaneously can cause damage or unpredictable behavior.

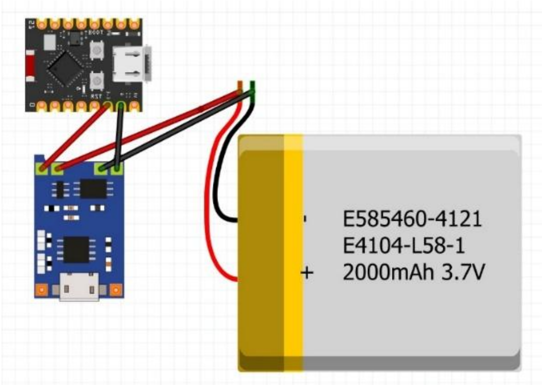


Figure 2.4 , Power supply circuit with TP4056 and LiPo battery.

At the measured average current draw of approximately 140 mA during active WiFi transmission, the 2500 mAh cell provides a theoretical runtime exceeding 17 hours, sufficient for a full overnight recording session with margin.

2.3 Sensors

The bracelet integrates five sensors, each measuring a different physiological signal. Their configuration parameters are summarized in Table 2.1. Each one is described in detail in the following subsections.

Table 2.1 , Sensor overview and key configuration parameters.

Sensor	Signal	Key parameters
MAX30102	PPG / SpO2	12.5 Hz effective, sampleAverage=8, sampleRate=200, ledBrightness=60, pulseWidth=411 µs
MPU6050	Accel / Gyro	±2 g / ±250 deg/s, boot calibration 300 samples
BMP280	Temperature / Pressure	Pressure ± 0.5 °C, temp SAMPLING_X2, pressure SAMPLING_X16, FILTER_X16
ADS1015 + Flex	Respiration	12-bit ADC, GAIN_SIXTEEN, baseline at startup (100 samples)
MAX4466	Snoring (audio)	200 Hz, 200-sample blocks, DC offset ~1066

2.3.1 MAX30102, PPG and Pulse Oximetry

The MAX30102 is an optical sensor that measures PPG and SpO2 by shining red (660 nm) and infrared (880 nm) light onto the skin and detecting the reflected signal [27].

The sensor runs at a hardware sample rate of 200 Hz with an averaging factor of 8, which gives a FIFO output of 25 Hz. The firmware polls the FIFO every 80 ms, but since each HTTP POST blocks the main loop for approximately 500 ms, the effective rate reaching the PPG buffer is 12.5 Hz. The pulse width is set to 411 µs, the longest available option, to maximize signal amplitude. The remaining parameters are adcRange=4096 and ledBrightness=60 out of 255. –

Figure 2.5 , MAX30102 sensor module (right) and positioned on the wrist during operation (left). The visible red illumination corresponds to the 660 nm LED; the infrared LED (880 nm) is not visible to the naked eye.

Wrist-worn optical measurement produces a lower signal-to-noise ratio compared to fingertip placement, a known limitation of wrist-mounted PPG sensors that is addressed in the bandpass filtering stage described in Chapter 4 [11].

2.3.2 MPU6050, Accelerometer and Gyroscope

The MPU6050 combines a three-axis accelerometer and a three-axis gyroscope on a single chip [28]. The accelerometer is configured at ±2 g, sufficient to capture the low-amplitude movements typical during sleep, and the gyroscope at ±250 deg/s for detecting body rotation and repositioning during the night.

The sensor is read at 10 Hz over I2C. At boot, a calibration routine collects 300 gyroscope samples at rest and computes the mean offset for each axis, which is subtracted from all subsequent readings. A five-point median filter is applied in firmware to suppress intermittent high-amplitude spikes caused by voltage fluctuations during WiFi transmission.

Figure 2.6 , MPU6050 inertial measurement unit module.

2.3.3 BMP280, Temperature and Pressure

The BMP280 is a low-power barometric pressure and temperature sensor with a specified temperature accuracy of ±0.5 °C over the range of -40 °C to +85 °C [29]. In this system it is used primarily for skin temperature estimation at the wrist.

The sensor runs in normal mode with temperature oversampling set to SAMPLING_X2 and pressure oversampling to SAMPLING_X16, a hardware IIR filter of FILTER_X16 to reduce short-term fluctuations, and a standby time of STANDBY_MS_500 between measurements. In practice this gives an output rate of approximately 1 Hz, which is appropriate since skin temperature changes slowly.

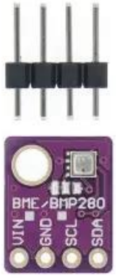


Figure 2.7 , BMP280 temperature and pressure sensor module.

2.3.4 ADS1015 + Flex Sensor , Respiratory Monitoring

Respiratory rate is monitored using a flex sensor mounted on an elastic band worn around the chest and secured with a Velcro closure. The flex sensor is a piezoresistive element whose resistance varies between approximately 10 kΩ and 110 kΩ depending on the degree of bending. Since the ESP32-C3 does not have an analog input with sufficient resolution for this signal, the flex sensor is connected to the ADS1015, a 12-bit I2C analog-to-digital converter configured with GAIN_SIXTEEN [30].

The sensor is mounted in a sandwich configuration with a coin acting as a rigid pivot point at the center, which amplifies the bending effect and produces a range of approximately 120 ADC units between full inspiration and full expiration. At startup, the firmware collects 100 samples at rest to establish a baseline, and all subsequent readings are stored as flex_delta = flex_raw minus baseline. The delta signal is bipolar, with positive values corresponding to inspiration and negative values to expiration, which allows respiratory rate to be extracted from the zero-crossing frequency in the analysis pipeline.

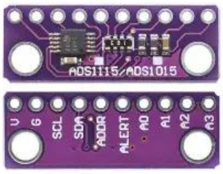


Figure 2.8 , ADS1015 12-bit analog-to-digital converter module.

Figure 2.9 , Flex sensor module.

The approach is low-cost and non-invasive, though the output is sensitive to sensor placement shifts and movement artifacts during overnight recordings.

2.3.5 MAX4466, Microphone for Snoring Detection

Snoring detection uses an electret microphone with an integrated MAX4466 amplifier [31]. The MAX4466 provides fixed gain and outputs a signal centered around DC bias. The microphone is sampled at 200 Hz in blocks of 200 samples, giving one block per second. Each block is included in the combined JSON packet sent to the server, where the raw samples are stored in the mic_data table.

The microphone is connected to GPIO0 of the ESP32-C3, which also functions as the boot selection pin. In normal operation this does not cause any issues, but it is worth noting for anyone modifying the firmware.

Figure 2.10 , MAX4466 microphone module.

2.4 PCB and Mechanical Integration

This section describes the physical assembly of the system, how the components are wired together and how the sensor assembly is attached to the body during a recording session.

2.4.1 Wiring and I2C Bus

All digital sensors share a single I2C bus with SDA on GPIO5 and SCL on GPIO6, running at 400 kHz (Fast Mode). Each sensor has a unique 7-bit address, so no multiplexer is needed. Address assignments are listed in Table 2.2. The microphone is the only exception, it connects directly to GPIO0 as an analog input

Table 2.2 , Sensor initialization notes and hardware configuration

Component	Notes
MPU6050	Register 0x6B: power management, write 0x00 to wake
BMP280	SDO pin pulled to GND
MAX30102	Library handles address internally
ADS1015	ADDR pin floating
Status LED	On-board LED for visual feedback

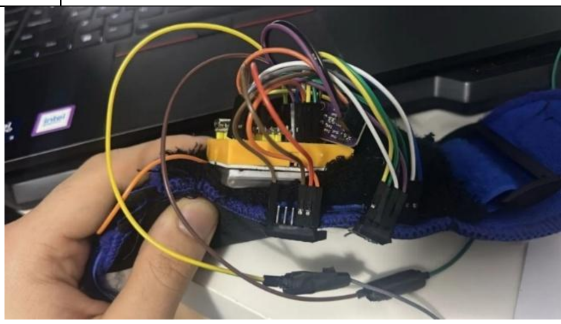


Figure 2.11 , Wrist unit with the tennis wristband open, showing the breadboard assembly, jumper wire connections, LiPo battery mounted underneath the breadboard, and the sensor modules.

2.4.2 Sensor Fixation on the Wrist

Mechanical fixation was the most persistent practical challenge during overnight recordings. The main issue is that the MAX30102 and BMP280 are sensitive to contact pressure and positioning and any shift during sleep directly affects signal quality. In several sessions the sensors moved during the night, which showed up as flat signal segments in the recorded data.

The current solution uses a tennis wristband as the main housing. The breadboard assembly sits inside the wristband, with the sensors sewn directly onto the inner fabric so that the BMP280 and MAX30102 maintain consistent contact with the skin. The wristband can be opened to access the USB port of the ESP32-C3, charge the battery, or make hardware modifications without disassembling the system. The flex sensor is mounted separately on an elastic chest band secured with a Velcro closure. For future sessions, a more rigid and form-fitting housing would help improve sensor contact further, along with better cable management to reduce mechanical stress on the jumper wire connections.

Figure 2.12 , Wrist unit worn during a recording session. The tennis wristband holds the breadboard assembly in place, with the MAX30102 and BMP280 sensors in contact with the skin on the inner side.

Figure 2.13 , Elastic chest band worn during a recording session, with the flex sensor and coin pivot point visible at the center. The band is secured around the torso to monitor respiratory movement.

Chapter 3. Data Acquisition Software

This chapter describes the software that runs on both devices in the system. It covers the ESP32-C3 firmware that reads the sensors and sends the data, the Raspberry Pi server that receives and stores everything in a SQLite database, the live dashboard used to monitor signals during an active session and a raw data quality analysis that runs after each recording to verify signal integrity before the sleep pipeline starts.

3.1 ESP32 Firmware

The firmware runs on the ESP32-C3 and handles everything from sensor initialization to data transmission. It is written in Arduino C++ with USB CDC on boot enabled, so the board can be programmed directly over USB without an external adapter. The main loop reads each sensor at its own rate and sends everything to the Raspberry Pi server every 2 seconds in a single combined HTTP POST.

3.1.1 Sensor Initialization and Calibration

At startup, the firmware first connects to WiFi and synchronizes the clock via NTP, then initializes the sensors in sequence: ADS1015, BMP280, MPU6050, and MAX30102. If any sensor fails to initialize, its status flag is set to false and it is simply skipped in the acquisition loop without stopping the rest of the system.

Two calibration routines run during initialization. After the ADS1015 is ready, the firmware waits 1.5 seconds and collects 100 flex sensor samples at rest to compute the baseline, all subsequent flex readings are reported as the difference from this value. After the MPU6050 initializes, a gyroscope calibration collects 300 samples over approximately 3 seconds while the device is stationary, computes the mean offset for each axis, and subtracts it from all future readings to compensate for the per-unit drift bias.

Once all sensors are ready, the firmware enters a polling loop and checks the /status endpoint on the Raspberry Pi every 10 seconds until the server confirms an active session. Only then does data acquisition start.

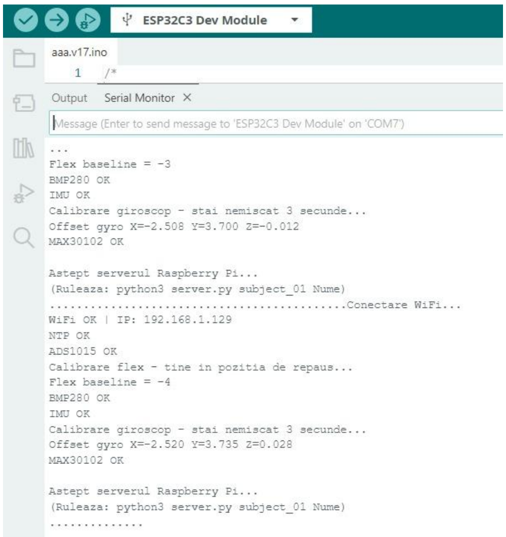


Figure 3.1 , ESP32 boot sequence with sensor initialization and calibration.

3.1.2 Sampling Frequencies per Sensor

Each sensor is read at a different rate, chosen based on how fast the signal changes and what the analysis pipeline needs. Table 3.1 summarizes the intervals and effective rates.

Table 3.1 , Sampling intervals and effective rates per sensor.

Sensor	Firmware interval	Effective rate	Notes
MAX30102 (PPG)	80 ms	12.5 Hz	sampleRate=200, sampleAverage=8, FIFO drain
MPU6050 (IMU)	100 ms	10 Hz	Direct I2C register read
ADS1015 (Flex)	250 ms	4 Hz	GAIN_SIXTEEN, baseline at boot
BMP280 (Temp)	1000 ms	1 Hz	Temperature changes slowly
MAX4466 (Mic)	5 ms	200 Hz	Blocks of 200 samples = 1 second
HTTP POST	2000 ms	0.5 Hz	Combined packet with all data

The MAX30102 hardware runs at 200 Hz with an averaging factor of 8, giving an effective FIFO output of 25 Hz. The firmware drains the FIFO every 80 ms, but since each HTTP POST blocks the main loop for approximately 500 ms, the effective rate reaching the PPG buffer is 12.5 Hz. The nextSample() call is critical, without it the FIFO pointer does not advance and the same stale sample repeats on every read.

The flex sensor is read at 4 Hz and buffered in the same way as PPG. Each reading computes flex_delta = flex_raw minus baseline, and both the raw and delta values are stored in separate buffers. The full block collected since the last send, typically around 8 samples, is included in the combined packet. This gives the pipeline enough resolution for respiratory rate extraction, which would not be possible with a single value per second.

All other sensors use simple timer-based polling. Gyroscope values are offset-corrected using the boot calibration and passed through a 5-point median filter to suppress intermittent spikes caused by voltage fluctuations during WiFi transmission.

3.1.3 JSON Data Format and HTTP POST

All sensor data is sent in a single combined HTTP POST to the /data endpoint every 2 seconds. This was a deliberate design choice driven by the fact that the ESP32-C3 is single-core. Each HTTP POST blocks the main loop for approximately 500 ms, so sending separate requests for sensors, PPG, flex, and microphone would cause significant sample loss in the buffers. The JSON payload contains the fields listed in Table 3.2 .

Table 3.2 , JSON payload fields in the combined /data POST.

JSON key	Source	Description
t	NTP clock	Sensor-side timestamp (HH:MM:SS)
fr, fd	ADS1015	flex_raw and flex_delta (raw − baseline)
ir, red	MAX30102	Latest IR and red values (single sample)
ax, ay, az	MPU6050	Accelerometer XYZ in g (4 decimal places)
gx, gy, gz	MPU6050	Gyroscope XYZ in deg/s (median filtered)

tmp, prs	BMP280	Temperature (°C) and pressure (hPa)
ppg_ir, ppg_red	MAX30102	Comma-separated PPG buffer (~12 samples)
fr_s, fd_s	ADS1015	Comma-separated flex buffer (~8 samples raw and delta)
mic_s	MAX4466	Comma-separated microphone block (200 samples)

The ppg_ir and ppg_red fields contain the PPG samples accumulated since the previous send, typically around 12 values. The fr_s and fd_s fields contain the flex buffer collected at 4 Hz since the last send, typically around 8 samples. The mic_s field contains the 200-sample microphone block if one is ready. If any of these buffers are empty at send time, the corresponding fields are simply omitted from the JSON. After a successful POST, all buffer indices are reset to zero and the microphone block flag is cleared.

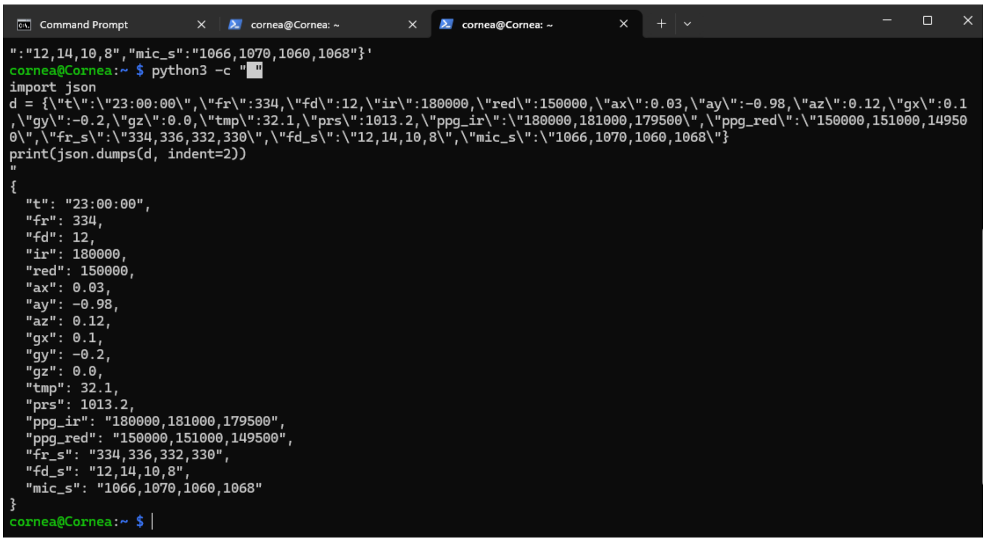


Figure 3.2 , Example JSON payload from a single combined POST.

3.1.4 Error Handling and LED Status Codes

The on-board LED on GPIO8 communicates the system state through a simple blink code. During the server polling phase before a session starts, the LED blinks slowly at 200 ms on and 800 ms off. Once data acquisition is active, each successful POST produces a brief 5 ms flash and a failed POST produces three rapid flashes of 50 ms each. If the server becomes unreachable during a session, the firmware pauses data transmission and polls the /status endpoint every 10 seconds until the server responds again, after which acquisition resumes automatically.

WiFi reconnection is not handled in the current firmware. If the connection drops, the HTTP POST fails and the error blink pattern appears, but the firmware does not attempt to reconnect. In practice the WiFi connection has been stable during overnight sessions, so this has not caused any issues.

3.2 Raspberry Pi Server

The Raspberry Pi acts as the data collection server, receiving POST requests from the ESP32 and storing everything in a SQLite database. It is implemented in Python using the

ThreadingHTTPServer class from the standard library, which handles incoming data and dashboard reads concurrently without one blocking the other.

3.2.1 HTTP Server Architecture

The server listens on port 5000 on all network interfaces, implemented in Python using ThreadingHTTPServer to handle incoming data and dashboard reads concurrently. A single-threaded server would block POST requests while servicing dashboard polls, so threading is necessary. The server exposes two endpoints: POST /data for the combined sensor packet, and GET /status which returns the current session state and record counts, used by the ESP32 to verify server availability before starting transmission.

HTTP logging is suppressed to keep the terminal output clean. Instead, the server prints a progress summary every 60 records showing the total raw rows, PPG blocks, flex blocks, and microphone blocks received so far.

3.2.2 Session Management and Session ID

Each recording session is started from the command line by running the server script with a subject identifier and a name. The session ID is generated automatically by combining the subject ID with the current date and time in the format subject_XX_YYYYMMDD_HHMMSS. This ID is attached to every record in the database, which makes it possible to have multiple sessions in the same file without any ambiguity. The session is stopped with Ctrl+C, which triggers a clean shutdown handler that prints the session summary including duration, total raw rows, PPG blocks, flex blocks, and microphone blocks, and suggests the pipeline commands to run for analysis.

3.2.3 SQLite Database, Tables raw_data, ppg_data, flex_data, and mic_data

All data collected during a recording session is stored in a SQLite database on the Raspberry Pi. SQLite was chosen for its simplicity, zero-configuration setup, and ability to handle concurrent reads and writes from multiple processes , the server writes incoming data, the dashboard reads it in real time, and the sleep pipeline reads it after the session ends, all from the same file.

The database uses five tables, each storing a different type of signal. Their relationships are shown in Figure 3.3.

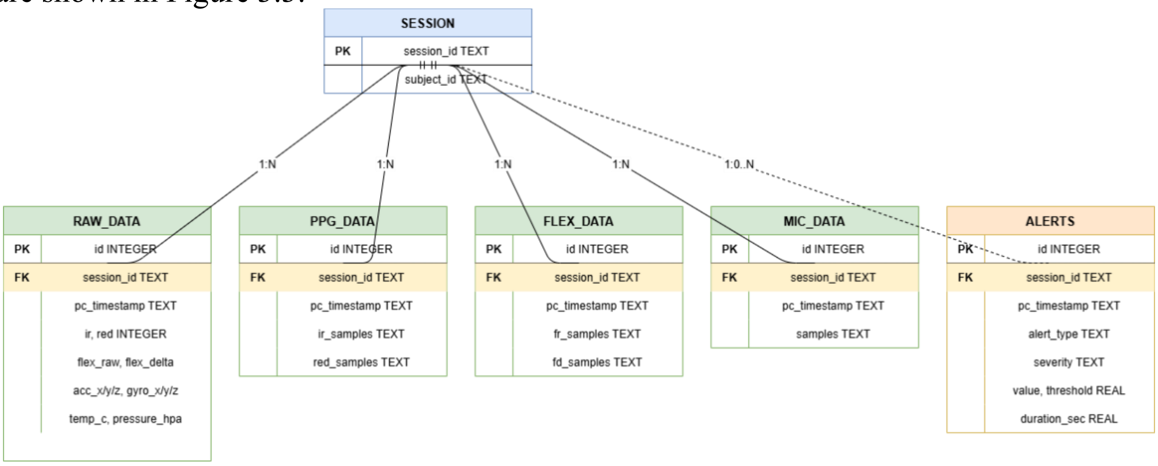


Figure 3.3 , SQLite database schema. SESSION acts as the logical parent entity linked to all data tables.

Table 3.3 summarizes the write frequency and content of each table.

Table 3.3 , Overview of all database tables.

Table	Frequency	Key columns	Description
raw_data	1 row / 2 s	flex_raw, flex_delta, ir, red, acc_x/y/z, gyro_x/y/z, temp_c, pressure_hpa, session_id	One snapshot per POST with the latest value from each sensor
ppg_data	1 block / 2 s	ir_samples, red_samples, session_id	PPG buffer (~12 samples per block as comma-separated strings)
flex_data	1 block / 2 s	fr_samples, fd_samples, session_id	Flex buffer (~8 samples per block as comma-separated strings)
mic_data	1 block / 1 s	samples, session_id	Microphone block (200 samples as comma-separated string)
alerts	per event	alert_type, severity, value, threshold, duration_sec, message, session_id	Real-time alerts generated by the alert monitor

WAL (Write-Ahead Logging) mode is enabled at database initialization, which allows the dashboard to read from the database without blocking concurrent server writes. Without WAL mode, a read operation locks the entire database file and incoming POST requests would fail during dashboard polling. All INSERT operations are wrapped in a threading lock with retry logic to handle the occasional "database is locked" error that occurs when multiple threads attempt to write simultaneously. Each retry waits progressively longer (100 ms, 200 ms, 300 ms) before giving up.

Indexes are created on (session_id, pc_timestamp) for each table at initialization. These are used by the dashboard's time-windowed queries, which request only the last 60 seconds of data on each poll cycle. Without the indexes, each query would perform a full table scan, at 1800 rows per hour for raw_data alone, this would become a bottleneck over a full overnight session.

3.2.4 Timeout Monitoring

The server includes a timeout mechanism that prints a warning if no data has been received for 120 seconds. This is implemented using a threading. Timer that is reset with each incoming record. If the timer expires, a message is printed to the terminal suggesting to check whether the ESP32 is still powered on and connected to WiFi. The timer then restarts itself so that repeated alerts are generated every 2 minutes until data resumes.

3.3 Live Dashboard

The live dashboard is a web application that displays all sensor signals in real time during a recording session. It runs on the Raspberry Pi on port 8080 and can be opened from any device on the local network. It runs as a separate process from the data server, so both can be active at the same time without interfering with each other. This is possible because the database uses WAL mode, which allows the dashboard to read data while the server writes new records concurrently. The main purpose is to verify during a session that all sensors are working correctly, the signal quality is acceptable, and the bracelet has not shifted on the wrist.

3.3.1 Architecture

The dashboard is implemented as a Flask application that serves a single HTML page with embedded JavaScript. The page is rendered once when the browser connects and then updated every 2 seconds by polling the /api/live endpoint. The endpoint queries the SQLite database for the last 60 seconds of data from all four tables (raw_data, ppg_data, flex_data, and mic_data), processes the signals server-side, and returns the results as a JSON object. The database is opened in read-only mode using SQLite URI mode with mode=ro, which prevents the dashboard from blocking the server writes. A retry mechanism with exponential backoff handles the occasional "database is locked" error, and a thread-safe cache avoids redundant database queries when multiple browser tabs are open at the same time.

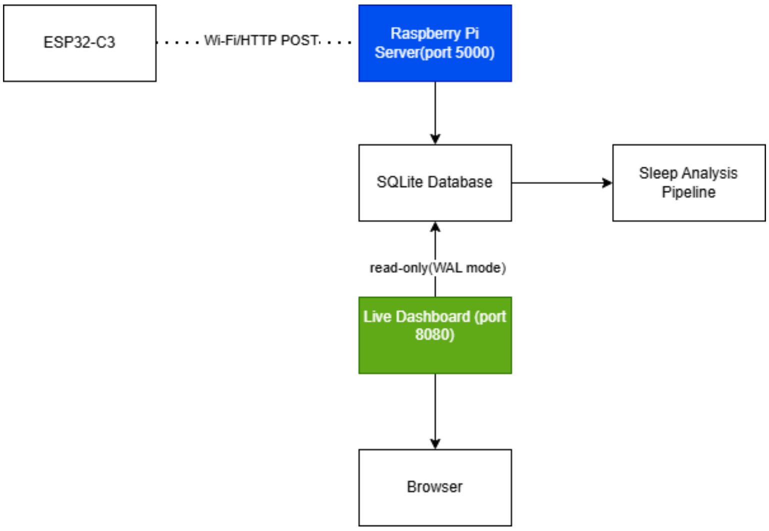


Figure 3.4 , Dashboard architecture showing the separation between data ingestion and visualization.

3.3.2 Signal Processing

All signal processing is done server-side in Python before the data reaches the browser. The frontend only handles chart rendering, which keeps the client lightweight and ensures that the signals displayed are consistent with what the analysis pipeline computes afterwards.
The PPG signal goes through two separate processing paths. For display, a Butterworth highpass filter at 0.5 Hz (order 2) is applied to remove the DC drift while keeping all harmonics above 0.5 Hz, which preserves the real pulse shape including the dicrotic notch. The signal is then normalized as AC/DC percentage and inverted so that cardiac peaks point upward following the standard PPG convention. For BPM estimation, the same raw samples go through a bandpass filter between 0.5 and 3.5 Hz instead, which removes both the DC baseline and the high-frequency noise that causes false peaks. BPM is computed via peak detection with a minimum distance of 0.5 seconds and a prominence threshold of 0.5 times the standard deviation, accepting only values between 40 and 120 BPM. The result is stabilized through a median filter over the last 7 detections. If scipy is not available, the dashboard falls back to a manual implementation that produces less accurate results but keeps the system functional.

The flex signal is smoothed with a 5-point moving average and centered on its median to isolate the respiratory component from the static baseline. The gyroscope values have their median subtracted to remove any residual offset not eliminated by the boot-time calibration. The microphone RMS is computed per block after mean subtraction to remove the DC bias.

3.3.3 Displayed Signals

The dashboard displays ten signals, each in its own chart or indicator card. Table 3.4lists all signals and the processing applied to each.

Table 3.4 , Dashboard signals and server-side processing.

Signal	Source sensor	Server-side processing
PPG Waveform	MAX30102	Highpass 0.5 Hz (Butterworth ord. 2), AC/DC normalization, inversion
BPM	MAX30102	Peak detection on bandpass 0.5–3.5 Hz, median smoothing (window=7)
Accelerometer	MPU6050	Raw (no filter needed)
Gyroscope	MPU6050	Median offset removal
Motion Level	MPU6050	Magnitude from acc and gyro axes
Flex / Respiration	ADS1015 + Flex	Moving average (window=5), centered on median
Temperature	BMP280	Raw value
Pressure	BMP280	Raw value
Microphone RMS	MAX4466	RMS per block after mean subtraction
Contact Quality	MAX30102	IR value with hysteresis debounce (3 frames), states: strong / weak / none

3.3.4 Contact Quality and BPM Stability

The dashboard includes a contact quality indicator based on the raw IR value from the MAX30102. Contact is classified into three states (strong, weak, and none) using a hysteresis mechanism that requires 3 consecutive frames in a new state before switching, which prevents rapid oscillation. When contact is lost, the PPG waveform and BPM are cleared immediately to avoid displaying stale data.

The BPM display uses a sticky mechanism: after a valid detection, the value stays on screen for up to 10 seconds even if no new valid reading occurs, preventing the number from flickering to zero during brief gaps caused by wrist movement. A confidence percentage shown alongside the BPM reflects how consistent the detected peak intervals are, a stable signal gives high confidence and an irregular one gives low confidence.

3.3.5 User Interface Layout

The interface is organized into three sections. The top row contains the BPM card, alongside six smaller vital sign cards showing PPG IR contact, temperature, flex delta, atmospheric pressure, microphone RMS, and the number of active samples. Each card has a sparkline mini-chart showing the last 60 seconds of the corresponding value and uses a color accent specific to that sensor type. Each card uses a distinct color accent specific to that sensor category.

Below the top row, a full-width panel displays the PPG waveform with badges showing the current BPM, the age of the last PPG block received, and the active filter type. The bottom section contains a grid of chart panels for the accelerometer (3 axes), gyroscope (3 axes, filtered), motion level (magnitude), and flex sensor (smoothed). Each panel can be clicked to open a full-screen overlay for closer inspection. A slide-out side panel accessible from the header shows the real-time status of each sensor with timestamps and raw values.

Figure 3.5 , Live dashboard interface during an active recording session.

Figure 3.6 , Full-screen PPG waveform overlay.

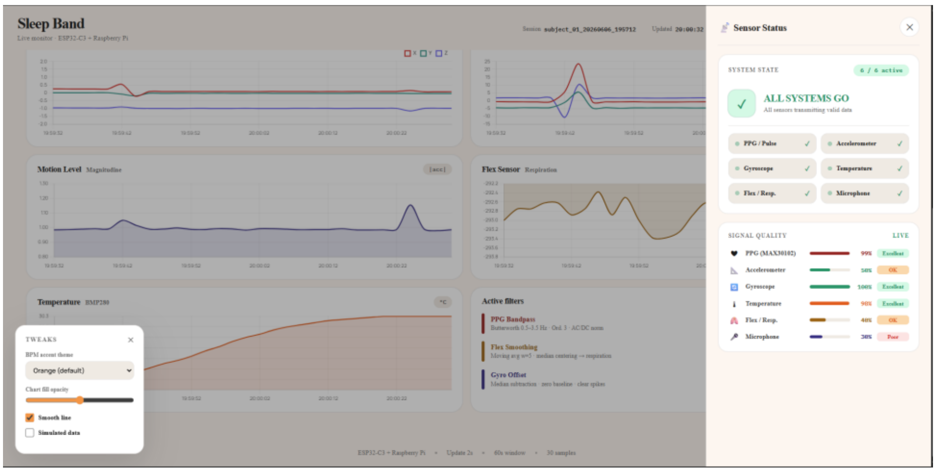


Figure 3.7 , Sensor status side panel

3.4 Raw Data Quality Analysis

After each recording session, the raw data is analyzed to assess signal quality before the sleep analysis pipeline runs. This step identifies periods of sensor detachment, excessive noise, or missing data that would affect the reliability of the features extracted in the next stage.

3.4.1 Signal Quality Metrics

For each sensor, the quality analysis computes a set of metrics over the entire session: SNR (signal-to-noise ratio), percentage of outliers defined as values more than 3 standard deviations from the mean, percentage of flat segments where consecutive values are identical, kurtosis as an indicator of non-Gaussian distributions, typically caused by motion artifacts or sensor spikes, and the percentage of missing values. Based on these metrics, each signal is assigned a status of OK, WARNING, or ERROR, which is printed in the analysis report and used to decide whether the session data is suitable for the sleep pipeline.

$$\text{outlier threshold} = \mu(\text{signal}) \pm 3 \times \sigma(\text{signal}) \qquad (3.1)$$

```
cornea@Cornea:~ $ python3 raw_data_analysis.py subject_10_20260519_000456
[LOAD] 17887 randuri | sesiune: subject_10_20260519_000456

============================================================
  RAPORT CALITATE DATE BRUTE – subject_10_20260519_000456
  Total randuri: 17887
============================================================

✅ IR PPG              | Status: OK          | SNR=  12.9 | outlieri=  1.1% | flat=  0.3%
✅ RED PPG             | Status: OK          | SNR=   6.2 | outlieri=  1.0% | flat=  0.6%
✅ AccX                | Status: OK          | SNR=   0.4 | outlieri=  2.6% | flat=  2.5%
✅ AccY                | Status: OK          | SNR=   0.6 | outlieri=  0.0% | flat=  2.5%
✅ AccZ                | Status: OK          | SNR=   2.0 | outlieri=  2.0% | flat=  1.9%
✅ GyroX               | Status: OK          | SNR=   0.1 | outlieri=  1.2% | flat=  1.7%
⚠️ Flex                | Status: ATENTIE     | SNR=   0.7 | outlieri=  0.0% | flat= 32.0%
   Probleme: semnal plat 32.0%
   Filtru recomandat: savgol
⚠️ Temperatura         | Status: ATENTIE     | SNR=  12.8 | outlieri=  0.0% | flat= 85.5%
   Probleme: semnal plat 85.5%
   Filtru recomandat: lowpass (None-0.05 Hz)

⚠️  Unele semnale necesita filtrare.
    Filtrele sunt aplicate automat in sleep_pipeline.py la extragerea features.
    Pentru date mai curate: verifica contactul senzorilor si conexiunile.

[PLOT] Grafic salvat in sleep_output/raw_quality_subject_10_20260519_000456.png
[PLOT] Salvat: sleep_output/raw_quality_subject_10_20260519_000456.png
```

Figure 3.8 , Signal quality report for a complete recording session.

In practice, the two signals most likely to show quality issues are the flex sensor and the BMP280 temperature. A high flat segment percentage on the flex signal almost always means the chest band shifted during the night and the sensor lost contact with the skin, rather than a genuine absence of respiratory movement. A high flat percentage on temperature, which in some sessions exceeded 85%, almost always means the BMP280 detached from the wrist. Both cases are flagged as WARNING in the report, and the corresponding periods should be excluded from the feature extraction or treated with caution during sleep stage classification.

3.4.2 Recommended Filters per Sensor

Based on the noise characteristics observed in the raw data, each signal has a recommended preprocessing filter applied in the analysis pipeline. The PPG signal uses a Butterworth bandpass filter between 0.5 and 3.5 Hz to isolate the cardiac component. The gyroscope applies median subtraction per window to remove residual offset without filtering the signal itself. The flex sensor uses a moving average with window 3 to smooth the respiratory waveform before peak detection.

Temperature uses a lowpass filter at 0.1 Hz to remove the high-frequency noise present in some sessions. These filters are described in detail in Chapter 4.

3.4.3 FFT Spectrum Analysis for PPG

The quality of the PPG signal was assessed in the frequency domain using Welch's periodogram with a Hanning window. The effective sampling rate of 5 Hz produces short usable signal segments, and a single-pass FFT on such a short signal yields high-variance spectral estimates where the noise floor fluctuates as much as the cardiac peak. Welch's method reduces this variance by dividing the signal into overlapping segments and averaging their individual spectra, which stabilises the noise floor and makes the dominant cardiac peak clearly identifiable. The Hanning window is applied to each segment before the FFT to suppress spectral leakage, without it, discontinuities at segment boundaries spread energy across adjacent frequencies and obscure the true peak location.

The spectrum is displayed over the 0.5 to 3.5 Hz cardiac band, with reference lines at 1.0 Hz and 1.5 Hz corresponding to 60 and 90 BPM. In a session with stable sensor contact, the spectrum contains a single dominant peak at the fundamental cardiac frequency. For session S15, this peak appears at 1.21 Hz, corresponding to 73 BPM, consistent with the time-domain estimate of 70 BPM obtained from peak detection on the same segment. A flat or multi-peaked spectrum indicates motion artefacts or sensor detachment, requiring additional validation before BPM values are considered reliable.

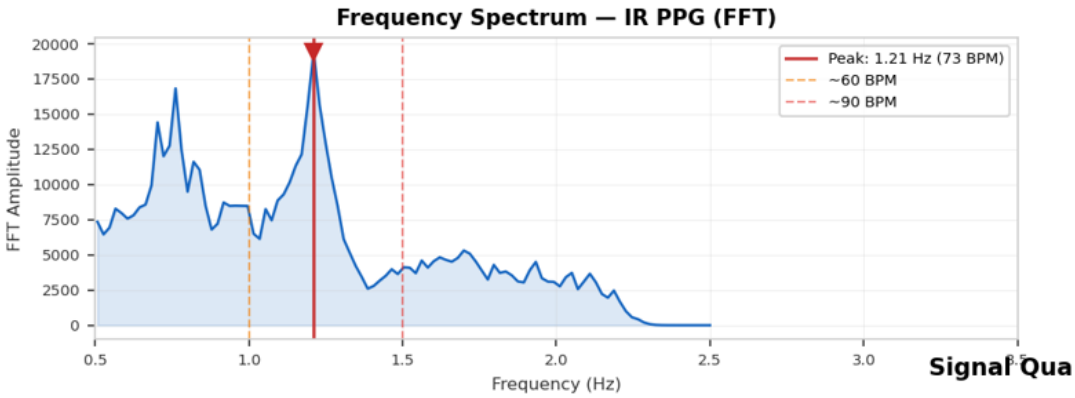


Figure 3.9 , Welch periodogram of the bandpass-filtered IR (PPG) signal over the cardiac band (0.5–3.5 Hz).

Chapter 4. Signal Processing and Feature Extraction

This chapter describes how the raw sensor data is processed and transformed into physiological features used for sleep stage analysis. It covers the preprocessing filters applied to each signal, the feature extraction from 30-second windows, and the normalization and weighting applied before the classifiers run.

4.1 Preprocessing and Filtering

Before any features are extracted, each signal is preprocessed to remove noise, artefacts, and non-physiological components. All filters are implemented using scipy.signal.filtfilt, which applies the filter forwards and backwards to achieve zero phase distortion. Table 4.1 summarizes the filter applied to each signal.

Table 4.1 , Preprocessing filters applied to each signal channel.

Signal	Filter type	Parameters	Purpose
PPG (IR)	Butterworth bandpass	0.5–3.5 Hz, order 3	Isolates cardiac component, removes DC and high-frequency noise
Accelerometer	None	—	Raw values used directly
Gyroscope	Median filter	kernel size 5	Suppresses spike artefacts, removes residual offset
Flex sensor	Moving average	window 3	Smooths respiratory waveform before peak detection
Temperature	Butterworth lowpass	0.1 Hz, order 3	Removes high-frequency noise from slow-changing signal
Microphone	Mean subtraction	per 200-sample block	Removes DC bias before RMS computation.

4.1.1 Bandpass Filter for PPG (0.5–3.5 Hz)

The PPG signal is filtered with a third-order Butterworth bandpass filter between 0.5 and 3.5 Hz. Before filtering, the signal is normalized as AC/DC percentage by subtracting the mean and dividing by it, which makes the filter parameters independent of absolute LED intensity.

$$signal\_normalized = (signal - mean) / mean \qquad (4.1)$$

The lower cutoff at 0.5 Hz removes DC baseline drift caused by changes in ambient light and sensor positioning. The upper cutoff at 3.5 Hz corresponds to a maximum heart rate of 210 BPM, well above any plausible range during sleep. When the effective sampling rate is low, the upper cutoff is capped automatically at 95% of the Nyquist frequency to avoid filter instability.

Figure 4.1 shows the IR signal deviation from the session mean across the full 486-minute recording for session S15. Flat regions near zero indicate stable sensor contact; abrupt steps correspond to movement or momentary sensor displacement. The signal quality metrics confirm SNR of 14.6 with 0.2% outliers, which is within acceptable bounds for overnight wrist PPG.

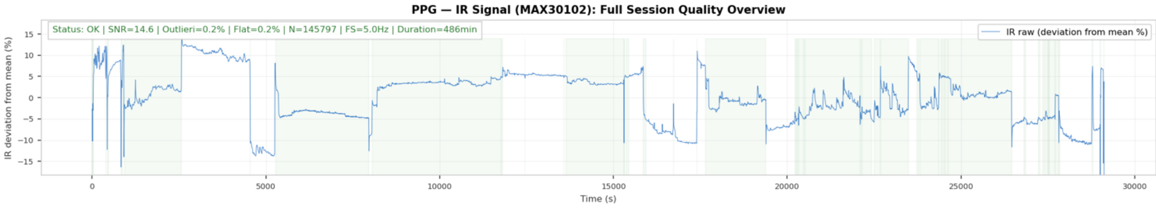


Figure 4.1 , Full session quality overview for session S15.

Figure 4.2 demonstrates the effect of the Butterworth bandpass filter on the raw PPG signal, captured in real time at 25 Hz using the Arduino Serial Plotter. The raw signal (blue) exhibits the cardiac pulse waveform with visible high-frequency noise and baseline fluctuations. The filtered output (orange), processed through a second-order Butterworth bandpass filter (0.5 to 3.5 Hz) implemented as two cascaded biquad sections on the ESP32-C3, retains the fundamental cardiac rhythm while attenuating noise and DC drift. The dicrotic notch is partially attenuated by the 3.5 Hz upper cutoff, as the notch morphology is encoded in the fourth and fifth harmonics of the cardiac frequency (4 to 6 Hz). Five consecutive cardiac cycles are visible, confirming stable sensor contact and consistent pulse detection.

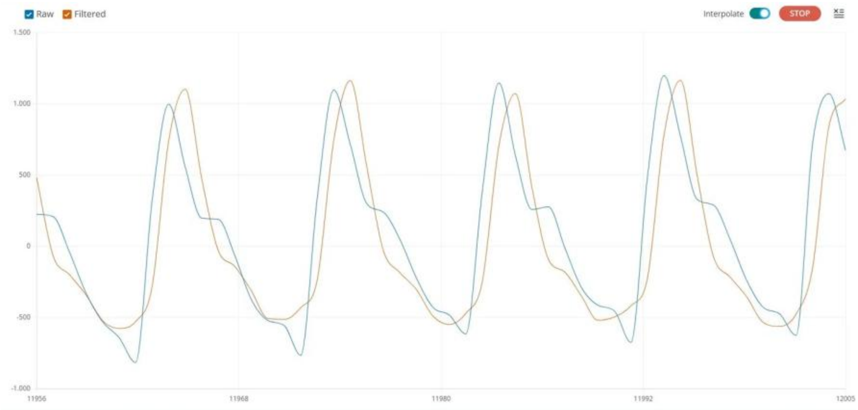
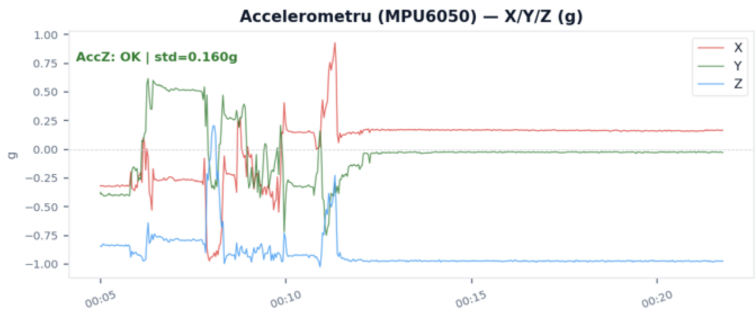


Figure 4.2 , Raw PPG signal (blue) and Butterworth bandpass-filtered output (orange) captured at 25 Hz via Serial Plotter.

4.1.2 Median Filter for Accelerometer and Gyroscope

The gyroscope signal is centered by subtracting the per-window median, which removes the residual offset without blurring the edges of real movement events. Intermittent high-amplitude spikes caused by voltage fluctuations on the power rail during WiFi transmission are suppressed by this approach, since they appear as isolated outliers relative to the local median. The accelerometer signal does not require additional filtering, the hardware averaging already applied by the MPU6050 at boot calibration is sufficient for the motion level computation used in feature extraction.



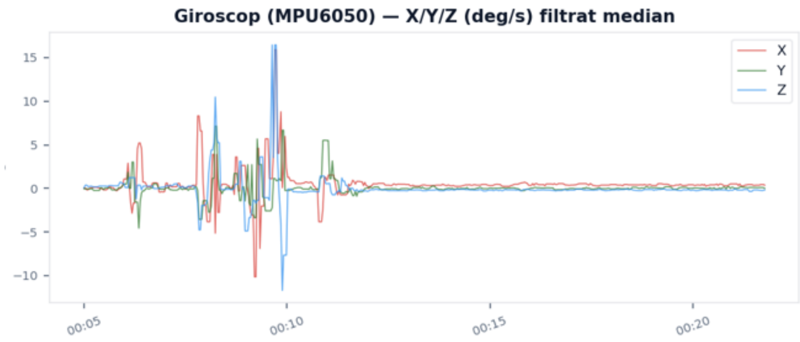


Figure 4.3 , Accelerometer and gyroscope signals from the MPU6050 during a recording session.

4.1.3 Moving Average Filter for Flex Sensor

The flex sensor signal is smoothed with a centered moving average of window 3 before peak detection. The filter reduces sample-to-sample noise while preserving the shape of the respiratory waveform, including peak amplitude and zero-crossing timing, both of which are needed for respiratory rate extraction. A window of 3 was chosen as the minimum that provides effective smoothing without introducing lag that would shift the detected breath timing.

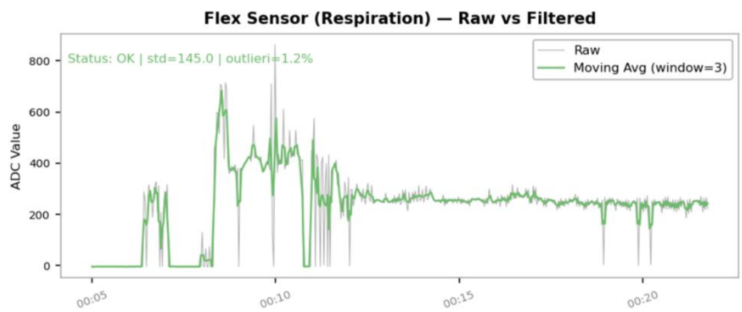


Figure 4.4 , Flex sensor respiratory signal before and after moving average filtering (window = 3).

4.1.4 Lowpass Filter for Temperature (0.1 Hz)

The BMP280 temperature signal is filtered with a third-order Butterworth lowpass filter at 0.1 Hz. Skin temperature changes very slowly during sleep, so any variation above 0.1 Hz is treated as noise. The filter smooths out the short-term fluctuations while keeping the slow circadian trend, which is the only physiologically relevant component in this signal.

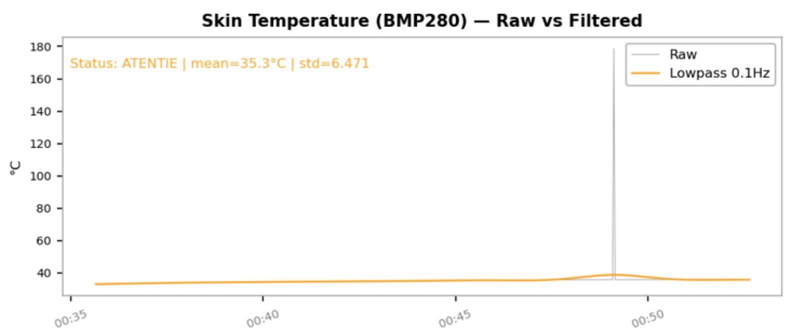


Figure 4.5 , Skin temperature signal before and after lowpass filtering at 0.1 Hz.

4.2 Feature Extraction, 30-Second Windows

The analysis pipeline splits the full recording into non-overlapping 30-second windows, following the epoch length used in clinical polysomnography. A window is included only if it contains at least 5 samples. For each window, features are extracted from all available sensor signals. When all sensors are functional the full set contains 21 features, but the pipeline handles missing or invalid data gracefully and works with fewer features when needed.

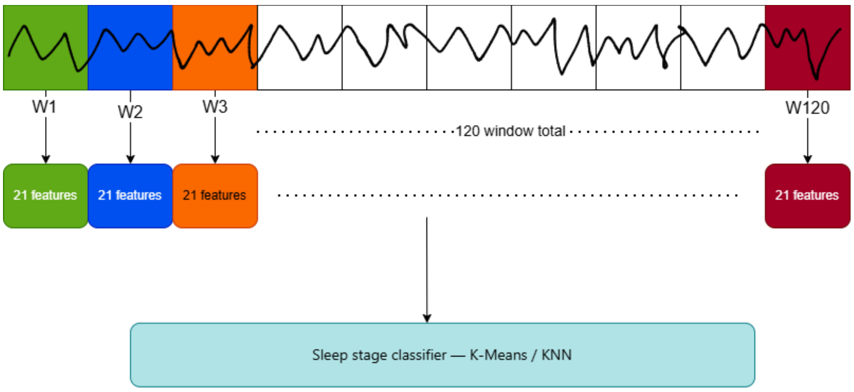


Figure 4.6 , 30-second window segmentation applied to the raw sensor data.

4.2.1 PPG Features: BPM, HRV (RMSSD), SpO2

PPG features are extracted from the high-rate PPG data in the ppg_data table when available. If not, the pipeline falls back to the IR and red values in raw_data. Before feature extraction, samples with IR values outside the 50000 to 250000 range are excluded as invalid contact, and a window is skipped entirely if fewer than 50% of samples pass this check. Table 4.2 lists the extracted PPG features.

Table 4.2 , PPG features extracted per 30-second window.

Feature	Unit	Description
bpm_mean	BPM	Mean heart rate over the 30-second window
bpm_std	BPM	Standard deviation of beat-to-beat heart rate
bpm_min	BPM	Minimum heart rate in the window
bpm_max	BPM	Maximum heart rate in the window
hrv_simple (RMSSD)	ms	Root mean square of successive RR differences after Malik filter (30% threshold)
spo2_estimated	%	SpO2 from ratio of ratios: 104 minus 17 times R (Maxim AN6409 calibration)
contact_valid_ratio	0–1	Proportion of samples with IR between 50000 and 250000 (valid contact)

Contact_valid_ratio is computed per window but excluded from the clustering and classification feature set. When the sensor is out of contact, this ratio is consistently low regardless of sleep stage. It reflects whether the hardware is attached, not what the body is doing. Including it would introduce data leakage into the classifier.

BPM is computed via peak detection on the bandpass-filtered signal with a minimum inter-peak distance of 0.5 seconds and a prominence threshold of 0.6 times the signal standard deviation, with an absolute minimum of 0.1. Only values between 40 and 120 BPM are accepted.

$$prominence_{min} = max(0.6 \times \sigma(signal), 0.1) \qquad (4.2)$$

HRV is computed as the RMSSD of the RR intervals after a Malik filter is applied. The Malik filter works by comparing each RR interval to its predecessor. If the difference exceeds 30% of the previous interval, that interval is considered an ectopic beat or a detection error and is removed before the RMSSD calculation. This step is important because wrist-based PPG peak detection is noisier than ECG, and a single false peak can introduce an artificially large RR difference that would inflate the RMSSD value significantly. The threshold was relaxed to 30% compared to the standard 20% used in clinical settings, to account for the higher variability of wrist PPG at low sampling rates.

$$RMSSD = \sqrt{[\,(1/N) \times \Sigma(RR_n - RR_{n-1})^2\,]} \quad (4.3)$$
$$|RR_n - RR_{n-1}| \,/\, RR_{n-1} > 30\% \qquad (4.4)$$

The SpO2 formula uses the Maxim AN6409 calibration, where R is the ratio of AC to DC amplitudes between the red and infrared channels, clipped to the 85 to 100% range.

$$SpO_2 = 104 - 17 \times R, \quad R = (AC_e^{id} \,/\, DC_e^{id}) \,/\, (AC^I_r \,/\, DC^I_r) \qquad (4.5)$$

4.2.2 Motion Features: Motion Level, Gyro Energy, Dominant Position

Six motion features are extracted from the accelerometer and gyroscope data per window. Table 4.3 lists them all. The most important is motion_level, where jerk is the first derivative of the acceleration magnitude. This formulation was chosen specifically because a stationary wrist has an accelerometer magnitude of approximately 1g due to gravity, computing motion as the deviation from 1g would miss the subtle micro-movements that distinguish light sleep from deep sleep. The standard deviation of the magnitude captures those micro-movements even when the wrist is nearly still. The acc_jerk term adds sensitivity to short rapid events like twitches and repositioning, which are physiologically relevant but brief enough to be averaged out by std alone.

$$motion\_level = std(acc\_magnitude) + 0.5 \times std(jerk) \qquad (4.6)$$

Table 4.3 , Motion features extracted per 30-second window.

Feature	Unit	Description
motion_level	g	std(acc_magnitude) + 0.5 * std(jerk) , captures micro-movements even when acc_mag ≈1g
motion_per_min	events/min	Number of jerk events (|diff(acc_mag)| > 0.01) per minute
motion_variation	dimensionless	Coefficient of variation of acc magnitude
acc_jerk	g/s	Standard deviation of the first derivative of acceleration magnitude
gyro_energy	deg/s	Standard deviation of centered gyroscope magnitude
dominant_position	0/1/2	Index of the axis carrying the gravity component (body orientation)

4.2.3 Respiratory Features: Respiratory Rate, Apnea Events

Five features are extracted from the flex sensor signal per window, listed in Table 4.4. Respiratory rate is computed using one methods selected based on signal quality. The method is peak detection on the smoothed flex_delta signal, with a minimum inter-peak distance of 2.5 seconds and a prominence threshold of 0.4 times the signal standard deviation. Intervals are converted to breaths per minute and filtered to the 3-40 range. If at least 3 valid peaks are found, this result is used directly.

$$prominence\_min = 0.4 \times std(flex\_delta) \tag{4.7}$$

Apnea events are detected using two complementary criteria combined with an OR rule: SpO2 desaturation of at least 4% below the session baseline confirmed over at least 2 consecutive windows, and respiratory amplitude falling below 15% of the session P75 baseline in windows where the flex sensor was confirmed active. Either criterion alone is sufficient to flag a window as an apnea event. A window is considered invalid for respiratory analysis if the peak-to-peak amplitude of flex_delta falls below 30 ADC units, which typically means the chest band has shifted during the night.

$$SpO_2 < SpO_2\_baseline - 4\% \tag{4.8}$$

$$resp\_amplitude < 0.15 \times P75(baseline\_amplitude) \tag{4.9}$$

Table 4.4 , Respiratory features extracted per 30-second window.

Feature	Unit	Description
resp_rate	breaths/min	Dominant respiratory frequency: peak detection when ≥3 valid peaks found
resp_variability	breaths/min	Standard deviation of inter-breath intervals
resp_amplitude	ADC units	Peak-to-peak range of flex_delta signal in the window
apnea_events_simple	count	Windows flagged by SpO2 desaturation >= 4% or flex amplitude < 15% of P75 baseline
breath_valid_ratio	0–1	Proportion of samples with movement above noise floor (10% of amplitude)

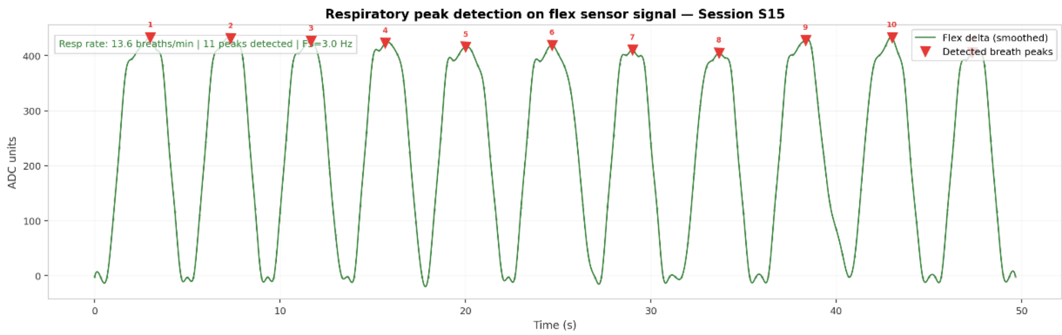


Figure 4.7 , Respiratory peak detection on the flex sensor signal over a 50-second window.

4.2.4 Temperature Features

Two features are extracted from the BMP280 temperature signal per window: mean skin temperature and its standard deviation. The mean provides a measure of absolute peripheral temperature, while the standard deviation captures short-term variability within the window. Temperature features are included in the feature set with a reduced weight because the

BMP280 frequently detaches from the wrist during overnight sessions, producing flat signal segments that are physiologically uninformative.

Table 4.5 , Temperature features extracted per 30-second window.

Feature	Unit	Description
temp_mean	°C	Mean skin temperature over the window
temp_variation	°C	Standard deviation of skin temperature

4.2.5 Acoustic Features: Snoring Detection

Snoring features are computed at session level rather than per window. For each 30-second window, the pipeline counts microphone blocks classified as snoring within a 60-second window centered on it. The matching window is intentionally wide because microphone blocks arrive approximately every 37 seconds, so not every analysis window has a directly corresponding block. The mean RMS of the snoring blocks found in that range is stored as snore_intensity.

$$RMS = \sqrt{\, mean(\, (x - \bar{x})^2 \,) } \qquad (4.10)$$

$$threshold = max(\, P97(all\_RMS),\ 3 \times median(all\_RMS),\ 150 \,) \qquad (4.11)$$

Table 4.6 , Acoustic features extracted per 30-second window.

Feature	Unit	Description
snore_events_count	count	Number of microphone blocks classified as snoring within a ±60s window centered on the epoch
snore_intensity	RMS	Mean RMS amplitude of the snoring blocks found in that window

4.3 Feature Weighting and Normalization

Before classification, all features go through three preprocessing steps: imputation of missing values, standardization using StandardScaler, and multiplication by physiological weights.

4.3.1 Feature Normalization

Missing values are handled differently depending on the feature type. For PPG features (BPM, SpO2), missing values are filled with zero rather than the session median. A NaN in these features almost always means the sensor was not in contact during that window, so zero is more informative than the median. HRV is treated separately: missing values are filled with the session median when valid HRV data exists, otherwise the column is excluded from the feature set entirely. For all other features, missing values are filled with the column median, which is robust to outliers. After imputation, all features are standardized to zero mean and unit variance using StandardScaler so that features with larger absolute values do not dominate the distance calculations in clustering.

4.3.2 Physiological Weighting Rationale

After standardization, each feature is multiplied by a physiological weight that reflects its expected discriminative power for sleep stage classification. Table 4.7 lists the weights. Motion features receive the highest weights (2.5) because body movement is the most reliable indicator of sleep stage with the sensors available in this system. BPM variability and HRV receive moderate weights because they are informative but noisier at the sampling rates achievable with wrist PPG. Absolute BPM receives a lower weight because heart rate does not vary dramatically across sleep stages in all subjects. Temperature is weighted at 0.8 because it frequently drifts or goes missing due to sensor detachment. The dominant position feature receives the lowest weight (0.3) since body orientation has limited relevance for sleep stage classification.

Table 4.7 , Physiological weights applied to each feature group.

Feature	Weight	Rationale
motion_level	2.5	Most reliable indicator for sleep staging
acc_jerk	2.5	Captures micro-movements, very discriminative
motion_per_min	2.0	Complementary motion indicator
bpm_std	1.8	BPM variability more informative than absolute value
resp_rate	1.8	Respiratory rate well correlated with sleep stage
hrv_simple	1.5	Reliable after Malik filter
gyro_energy	1.5	Complementary to accelerometer
resp_variability	1.5	REM respiratory irregularity
spo2_estimated	1.2	Moderate contributor
bpm_mean / min / max	1.0	Absolute BPM is a weak stage indicator
resp_amplitude	1.0	Peak-to-peak flex range; used for apnea threshold baseline
apnea_events_simple	1.0	Direct indicator of respiratory disruption
temp_mean	0.8	Temperature drifts, limited reliability
breath_valid_ratio	0.5	Quality flag; low discriminative power on its own
temp_variation	0.5	Short-term temperature variance; rarely informative
motion_variation	0.5	Complementary motion indicator
dominant_position	0.3	Not relevant for stage classification

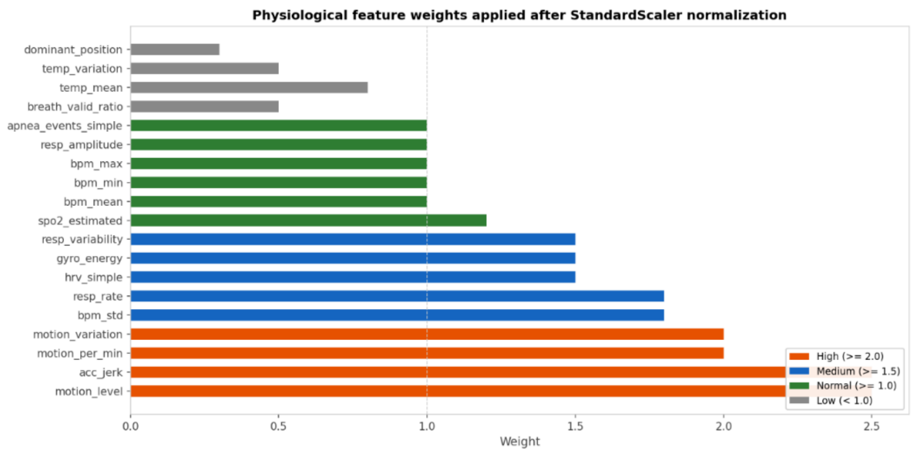


Figure 4.8 , Feature weights used in the classification pipeline.

Chapter 5. Sleep Stage Classification

The extracted features are used to classify each 30-second window into one of four sleep stages. The pipeline applies K-Means clustering to find natural groupings without requiring predefined labels, followed by a physiological labeling step that assigns a sleep stage to each cluster. A KNN classifier is then trained on those labels to produce per-window stage probabilities. The chapter also covers the sleep quality score and the PCA visualization used to assess stage separation in the feature space.

5.1 K-Means Clustering

K-Means is an unsupervised clustering algorithm that partitions the feature space into k groups by minimizing within-cluster variance [34]. It does not require predefined labels, which makes it suitable here since there is no ground truth annotation for the recorded sessions. The algorithm works by iteratively assigning each window to the nearest cluster centroid and updating the centroids until assignments no longer change.

5.1.1 Algorithm Description and Configuration

The pipeline uses KMeans from scikit-learn [36] with k = 4 clusters, corresponding to Light Sleep, Deep Sleep, REM Sleep, and Wake. When the cluster count is fixed, the model runs with n_init = 20 random initializations and max_iter = 500 iterations. When automatic silhouette-based selection is active, each candidate k is tested with n_init = 10 and max_iter = 300 to keep the search fast [37]. In both cases random_state = 42 is set for reproducibility.

The algorithm is fitted only on windows with valid PPG contact, defined as contact_valid_ratio >= 0.5. Windows where the sensor was not in contact are excluded from the fit to avoid pulling cluster centroids toward non-physiological values. After fitting, those windows receive the label Invalid / No Contact and are excluded from all subsequent analyses including the sleep quality score.

5.1.2 Automatic Cluster Count Selection via Silhouette Score

To evaluate how well the four-cluster solution fits the data, the silhouette score was computed for values of k from 2 to 6 [37]. The silhouette score measures how similar each window is to its own cluster compared to the nearest other cluster, ranging from -1 (poor separation) to 1 (well separated).

The results show that k = 2 produces the highest score (0.797), while k = 4 scores only 0.290, below the 0.5 threshold generally considered acceptable [37]. This is physiologically interpretable: the two natural groupings in the data correspond to active periods with high motion and irregular cardiac activity, and quiet periods with low motion and regular cardiac activity, mapping roughly to Wake versus Sleep rather than four distinct stages. The four-stage separation that PSG achieves relies heavily on EEG signals not available in this system.

The pipeline uses k = 4 regardless, because the four-stage model is the standard clinical framework and produces more interpretable output for comparison with published sleep research [3]. The low silhouette score at k = 4 is an acknowledged limitation, confirming that clustering alone is not sufficient for reliable stage classification. This is precisely why the physiological labeling step and the KNN classifier are applied afterwards to refine the initial groupings.

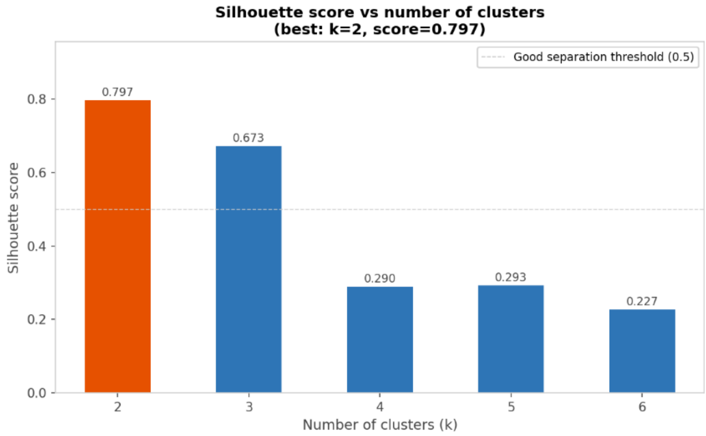


Figure 5.1 , Silhouette score for different values of k (number of clusters).

5.1.3 Sleep Stage Labeling, Combined Physiological Score

After clustering, the pipeline assigns sleep stage labels in two phases. In the first phase, the cluster with the highest median motion_level is labeled Wake / Movement, and all other clusters are initially marked as SleepCandidate. In the second phase, each SleepCandidate window is reclassified individually based on physiological rules applied to its own feature values. Table 5.1 summarizes the rules.

Table 5.1 , Physiological rules used for per-window sleep stage labeling.

Stage	Primary rule	Secondary condition
Wake / Movement	motion_level >= P95 of sleep windows	AND motion_variation > 0.01
REM Sleep	bpm_mean >= P65 AND bpm_std >= P55	Both conditions required
Deep Sleep	bpm_mean <= P40 AND motion_level <= P50	AND bpm_std <= P45 (capped at 25% of total)
Light Sleep	All remaining windows	Default category

Two physiological caps are applied after labeling. Deep Sleep is capped at 25% of total sleep windows, since N3 typically accounts for 20-25% of total sleep time in healthy adults [3]. If more windows are classified as Deep than this threshold allows, the excess windows with the highest motion_level within the Deep group are reassigned to Light Sleep. REM Sleep is capped at 30% of total sleep windows for the same reason [4]. Both thresholds reflect the known distribution of sleep stages and prevent the rule-based labeling from over-assigning either stage in sessions where the subject was unusually still or had consistently elevated heart rate variability throughout the night.

5.1.4 Hypnogram Generation and Smoothing

After labeling, a majority-vote smoothing filter with a window of 3 epochs (90 seconds) is applied to remove isolated transitions that are likely classification noise rather than real stage changes. The filter replaces each window's label with the most common label in its 3-window neighborhood, while leaving Invalid / No Contact labels unchanged. The original unsmoothed labels are kept in the sleep_stage_raw column for reference. The smoothed labels are then used to generate the hypnogram, which shows the sleep stage progression across the full recording plotted against clock time.

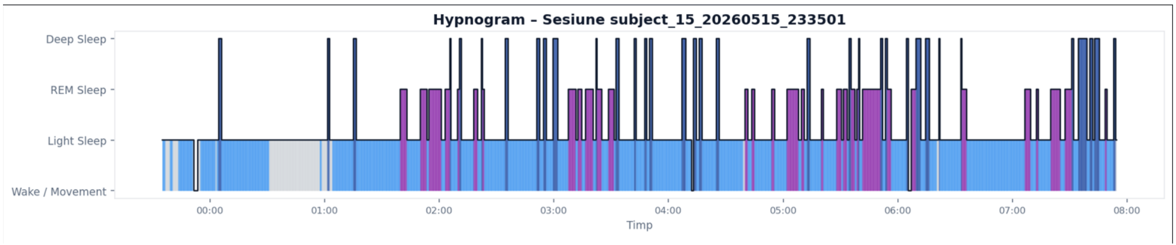


Figure 5.2 , Hypnogram generated from K-Means classification with smoothing applied.

The hypnogram produced by this system exhibits frequent narrow transitions, particularly for Deep Sleep and REM Sleep, appearing as thin vertical spikes. This fragmentation is inherent to automatic per-epoch classification: at an 8-hour recording scale, even physiologically valid blocks of 2 to 3 consecutive epochs (60 to 90 seconds) appear as narrow lines. This is consistent with published wearable-based sleep staging systems, which typically produce more fragmented hypnograms than polysomnography [11], [15].

5.2 K-Nearest Neighbors (KNN)

After K-Means produces the initial stage labels, a K-Nearest Neighbors classifier is trained on the same feature vectors using those labels as the target [35]. The purpose of KNN is not to replace K-Means but to complement it. KNN provides a per-class probability for each window, which gives a measure of how confident the classification is, and it tends to produce smoother stage boundaries than the hard cluster assignments from K-Means because it classifies each window based on its similarity to nearby training examples rather than its distance to a centroid.

The classifier uses k = 5 neighbors with Euclidean distance and distance-based weighting, so closer neighbors contribute more to the final prediction than distant ones [36]. The model is trained on the full session and outputs both a predicted label stored in knn_stage and per-class probabilities for each of the four sleep stages.

5.2.1 Algorithm Description and Configuration

The KNN classifier is configured with k = 5 neighbors, Euclidean distance in the 21-dimensional weighted feature space, and distance-based voting where closer neighbors have proportionally more influence on the final classification [35]. Table 5.2 summarizes the full configuration.

Table 5.2 , KNN classifier configuration.

Parameter	Value	Description
n_neighbors (k)	5	Number of nearest neighbors used for classification
metric	euclidean	Distance computed in the 21-dimensional weighted feature space
weights	distance	Closer neighbors have proportionally more influence on the vote
Input X	X_scaled × weight_arr	Standardized and weighted feature vectors (21 dimensions)
Input y	sleep_stage (K-Means)	Training labels derived from the K-Means labeling step
Output	knn_stage + probabilities	Predicted stage plus per-class probability for each window

Distance-weighted voting is particularly useful at stage boundaries, where a window is physiologically ambiguous between two stages. Instead of treating all five neighbors equally, the classifier gives more weight to the ones most similar to the window being classified, which produces more accurate predictions in the transition zones between Light Sleep and REM Sleep [35].

5.2.2 Training on K-Means Labels

The KNN classifier is trained on the full set of windows using the sleep stage labels produced by K-Means as the target, making it a supervised classifier built on top of an unsupervised foundation [35]. The implicit assumption is that the K-Means labels, after the physiological labeling step, are a reasonable approximation of the true sleep stages. The KNN then learns to reproduce those labels from the feature vectors and assigns a stage to any window by finding its nearest neighbors in the training set.

One consequence of this approach is that the KNN cannot correct systematic errors in the K-Means labels. If the clustering misclassifies a group of windows, the KNN will learn to reproduce those mistakes. This is an acknowledged limitation of the pipeline, and it is one reason why the classification results are validated against subjective morning questionnaire responses rather than treated as ground truth [8].

5.2.3 Per-Class Probability Output

For each window, the KNN produces a probability for each of the four sleep stages: prob_Deep_Sleep, prob_Light_Sleep, prob_REM_Sleep, and prob_Wake_Movement. These probabilities reflect the proportion of the k nearest neighbors belonging to each class, weighted by distance, so they always sum to 1. A window classified as REM Sleep with prob_REM_Sleep = 0.8 is a confident classification. One with prob_REM_Sleep = 0.4 and prob_Light_Sleep = 0.4 indicates ambiguity at a stage boundary. These probabilities are stored in the output CSV alongside the predicted stage and can be used to identify windows where the classification is uncertain.

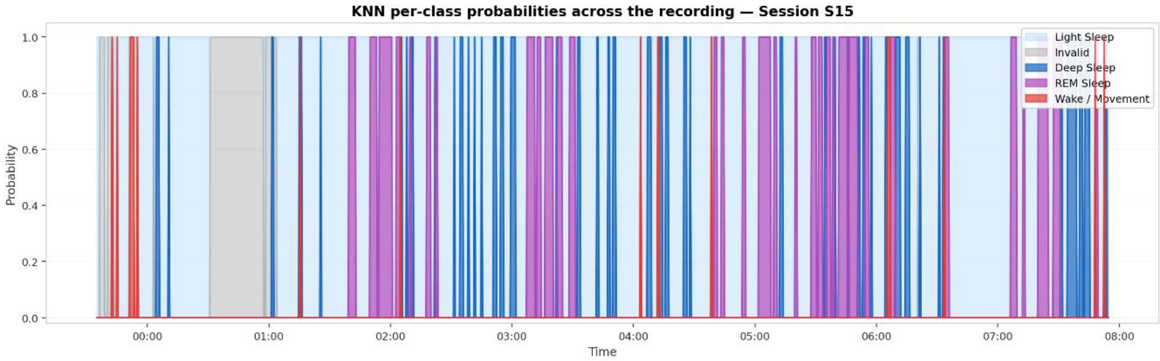


Figure 5.3 , KNN per-class probabilities across the recording.

Figure 5.3 shows that Light Sleep dominates the recording with consistently high probability, reflecting its 68% share of total sleep time. REM Sleep probability peaks are visible mainly after 02:00, consistent with the known pattern of REM-rich sleep in the second half of the night [6]. Transitions between Light Sleep and REM Sleep show partial probability overlap, indicating classification uncertainty at stage boundaries. Deep Sleep and Wake probabilities remain low throughout, appearing only as brief spikes.

The near-binary appearance of the probability traces is a known consequence of training and evaluating KNN on the same dataset. When a window's nearest neighbors are identical to the training examples, the classifier assigns probability 1.0 to one class and 0.0 to the others. This effect would be reduced if the classifier were evaluated on an independent held-out set, but in the absence of ground truth annotations this is not possible within the current pipeline.

5.3 Sleep Quality Score (0–100)

After sleep stage classification is complete, the pipeline computes a single numerical score between 0 and 100 that summarizes the physiological quality of the recording session. Rather than reporting only a stage distribution, the score aggregates six distinct components derived from the extracted features, each weighted according to its clinical relevance to sleep quality [3]. Windows classified as Invalid / No Contact, meaning windows where the PPG sensor lost skin contact, are excluded from all score calculations. The valid recording duration used for scoring is therefore the total session length minus any disconnected intervals. The final score is clipped to the 0 to 100 range, with an additional duration penalty applied for sessions shorter than 5 hours, and mapped to a qualitative grade.

5.3.1 Score Components and Weights

Table 5.3 lists the six components, their maximum point values, the optimal physiological ranges, and the scoring logic applied to each. The total maximum is 100 points before any duration penalty is applied.

Table 5.3 , Sleep quality score components and scoring formula.

Component	Max points	Optimal range	Scoring logic
Sleep architecture	30	Deep≥25%, REM≥25%, Wake<15%	10 pt per Deep/REM up to target; penalty for Wake above 15%
PPG quality (BPM + SpO2)	20	BPM 45–70, SpO2≥95%	10 pt BPM in range, 10 pt SpO2 ≥95%
Respiration	20	RR 10–18, apnea <2% of windows	10 pt RR in range (6 pt if 8–22); apnea: 10/7/4/1 pt for <2%/<5%/<15%/>15% of windows
Movement	15	motion < 0.05 g	15/10/5/0 pt for median motion <0.05/<0.15/<0.30/>0.30 g
HRV	10	30–100 ms	10/6/3 pt for 30–100ms/15–150ms/outside range; 5 pt if HRV unavailable
Snoring	5	0 events	5/4/3/2/1 pt for 0/1–20/21–50/51–100/>100 total snoring events
Duration penalty	up to −15	≥6 hours	Fixed −15 pt below 4 hours; linear reduction between 4 and 6 hours

The architecture component carries the highest weight (30 points) because the ratio of Deep and REM sleep is the most clinically meaningful indicator of sleep quality [3]. The Wake penalty operates on a linear scale rather than a threshold: a session with no movement epochs scores the full 10 points, and the score reaches zero at 15% Wake or above. The respiration component matches the PPG component in weight because respiratory rate and apnea events are direct physiological measurements, while BPM and SpO2 at the wrist are noisier at the sampling rates used [17]. The HRV component defaults to 5 points when the PPG sampling rate is too low for reliable extraction, rather than penalizing sessions where the measurement was not possible. The snoring component has the lowest weight (5 points) because the microphone-based detection has a higher false positive rate than the other sensors.

5.3.2 Duration Penalty

A duration penalty is applied when the valid recording duration is below 6 hours. Below 4 hours the penalty is a fixed 15 points. Between 4 and 6 hours the penalty scales linearly from 15 to 0 points. Above 6 hours there is no penalty. The penalty is applied to the total score after all component scores are summed, and the result is clipped to the 0 to 100 range. The duration used is the valid recording time, excluding windows marked as Invalid / No Contact. The 6-hour threshold reflects the minimum sleep duration associated with adequate physiological recovery in adults [7].

$$
\begin{cases} 15 \text{ pt}, & \text{if duration} < 4\text{ h} \\ 15 \times (6 - \text{hours}) / 2 \text{ pt}, & \text{if } 4\text{ h} \leq \text{duration} \leq 6\text{ h} \\ 0 \text{ pt}, & \text{if duration} > 6\text{ h} \end{cases} \quad (5.1)
$$

5.3.3 Grade Classification

The final score is mapped to a qualitative grade as shown in Table 5.4 .The thresholds were chosen to align with the physiological ranges used in the scoring components: a session scoring 85 or above has most components within optimal ranges, while a session below 40 has either multiple degraded components or a recording too short to be meaningful.

Table 5.4 , Sleep quality score grade thresholds.

Score range	Grade	Interpretation
85–100	Excellent	All physiological components within optimal ranges
70–84	Good	Most components normal, minor deviations
55–69	Average	Some components outside optimal range
40–54	Poor	Multiple components degraded
0–39	Very poor	Significant physiological anomalies or very short recording

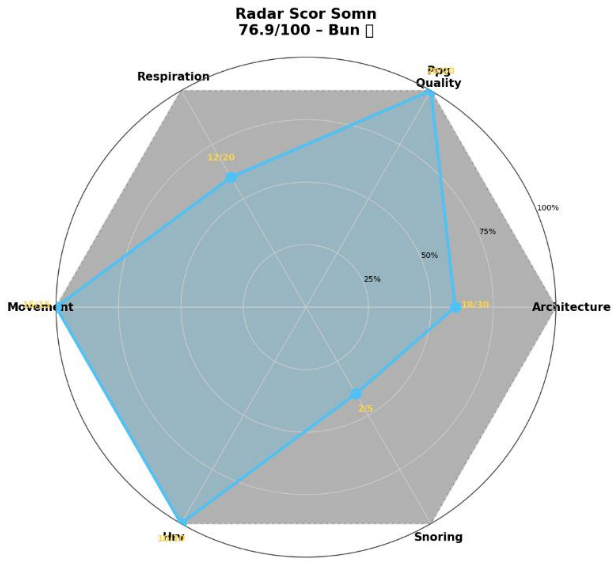


Figure 5.4 , Sleep quality score component breakdown.

5.4 PCA Visualization

Principal Component Analysis is applied to the scaled and weighted feature matrix to produce a 2-dimensional representation of the full feature space (18 to 21 dimensions depending on sensor availability). The PCA is used only for visualization, not for classification. Its purpose is to give an intuitive view of how well the four sleep stages separate in the feature space [36].

5.4.1 Dimensionality Reduction from 21 to 2 Dimensions

PCA is implemented using PCA(n_components=2, random_state=42) from scikit-learn [36], fitted on the full set of valid windows for each session. The first two principal components capture the directions of maximum variance in the feature space. The percentage of total variance explained by each component is shown on the axis labels.

5.4.2 Cluster Separation Analysis

Each window is plotted as a point in the 2D PCA space, colored by its sleep stage label. Well-separated clusters indicate that the features carry enough information to distinguish the sleep stages; overlapping clusters indicate ambiguity. The axis limits are set to the 1st–99th percentile range on each axis, with a 10% margin added on each side. Points outside this range are still rendered but fall outside the visible area, their count is reported in the plot title. This prevents extreme values from compressing the main cluster area without discarding any data.

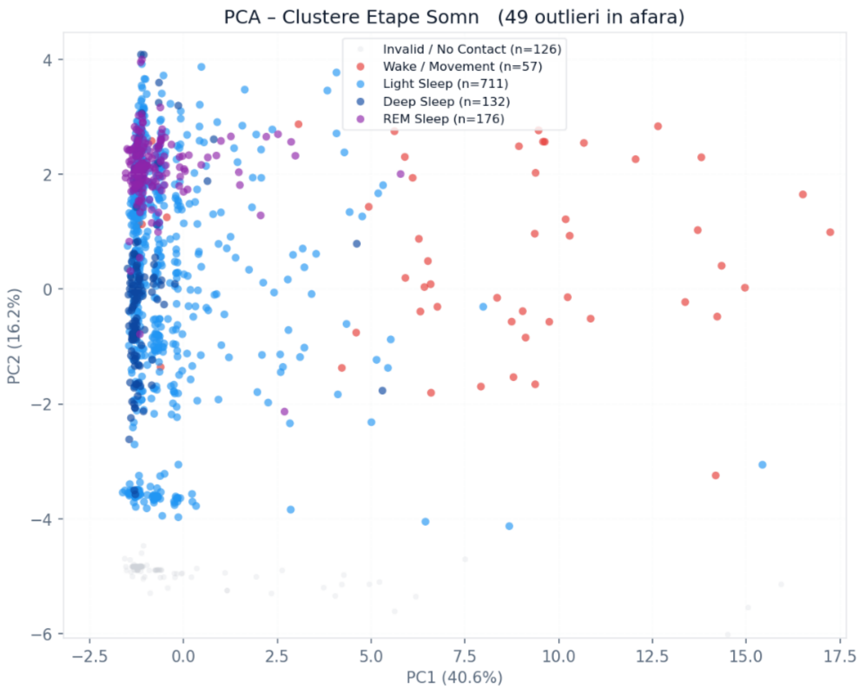


Figure 5.5 , PCA visualization of sleep stage clusters in the reduced feature space.

PC1 (40.6%) separates Wake / Movement clearly from the sleep stages, which is expected given that motion features carry the highest preprocessing weights in the pipeline [33]. Light Sleep, Deep Sleep, and REM Sleep form an overlapping cluster in the left region of the plot. The three stages produce similar values on the PPG and motion features available in this session, and the features that best distinguish them physiologically, HRV and respiratory rate, contributed less to the classification given their lower physiological weights and higher rates of missing data. The overlap in PCA space reflects the same classification uncertainty visible at the Light Sleep / REM Sleep boundary in the probability output, and confirms that 2D projection captures the main limitation of the current feature set rather than a failure of the clustering algorithm.

Chapter 6. Results and Validation

This chapter presents the experimental results obtained from ten overnight recording sessions collected from eight subjects. It covers the raw signal quality observed across sessions, the sleep stage classification outputs produced by the K-Means and KNN pipeline, the sleep quality scores, and a comparison between the objective sensor measurements and the subjects' own self-reported sleep quality from the morning questionnaire.

6.1 Experimental Setup

6.1.1 Subject and Session Description

The system was tested on eight subjects from the immediate social and family environment, producing ten overnight recording sessions in total. Two subjects were each recorded on two separate nights: one subject contributed sessions S09 and S16, and another contributed sessions S10b and S28, providing a limited basis for assessing night-to-night variability within the same individual. No participant had a diagnosed sleep disorder at the time of recording, although the pipeline detected respiratory events in several sessions that may indicate subclinical apnea activity. Sessions collected during hardware development and short calibration recordings under 30 minutes were excluded from the main analysis.

Subject ages ranged from 20 to 70 years. The group included both male and female participants. The questionnaire collected demographic and lifestyle data alongside the subjective sleep quality ratings, but the sample size is too small to draw demographically stratified conclusions, so the analysis treats all subjects as a single group.

6.1.2 Data Collection Protocol

Each recording session began after the subject fitted the wrist unit and chest band in the position described in Chapter 2. The server was started on the Raspberry Pi before the subject went to sleep, and the ESP32 firmware connected automatically once it detected an active session. The subject went to sleep normally without any constraints on sleep time, wake time, or sleeping position.

In the morning, each subject completed a sleep quality questionnaire via Google Forms [8]. The form collected bedtime and wake time, subjective sleep quality and feeling of refreshment on a 1–5 scale, number of nocturnal awakenings, perceived movement during sleep, pre-sleep stress level, reported snoring, and lifestyle factors including caffeine consumption, alcohol, and screen time before sleep. Responses were matched to their corresponding recording session by session ID for the correlation analysis in Section 6.4.

Recording duration varied across subjects depending on individual sleep schedules, ranging from 375 to 602 minutes with a mean of approximately 496 minutes. Valid PPG coverage ranged from 0.72 to 1.00 across sessions. Table 6.1 summarizes all ten sessions.

Table 6.1 , Overview of the ten recording sessions included in the experiment.

Session ID	Total Duration (min)	Valid PPG Duration (min)	Valid PPG Ratio	Deep (%)	Light (%)	REM (%)	Wake (%)	Sleep Quality Score
S05	494	494	N/A*	53.5	0.1	46.3	0.1	68.9
S06	436	436	N/A*	21.7	14.7	62.3	1.4	73.8
S09	538	388	0.72	13.5	58.5	25.0	3.0	80.4
S10a	557	461	0.83	21.6	65.8	7.7	4.9	65.5
S10b	602	602	1.00	2.2	92.2	1.7	4.0	73.9
S12	375	342	0.91	8.2	51.7	37.2	2.9	62.8
S15	500	459	0.92	11.3	68.2	18.4	2.1	76.5
S16	601	538	0.90	12.3	66.1	16.4	5.3	76.9
S17	448	446	1.00	9.9	71.2	15.2	3.7	76.6
S28	497	497	1.00	5.7	80.1	9.4	4.8	74.8

Sessions S05 and S06 were collected early in the study using an earlier version of the analysis pipeline that did not yet include the physiological cap mechanisms for Deep Sleep and REM Sleep, the valid PPG contact filtering, or several of the signal quality improvements described in Chapters 3 and 4. As a result, their stage distributions fall outside the ranges produced by the current pipeline and should be interpreted as preliminary results from an earlier development stage rather than as representative outputs of the final system. They are included for completeness to document the evolution of the pipeline across the study period.

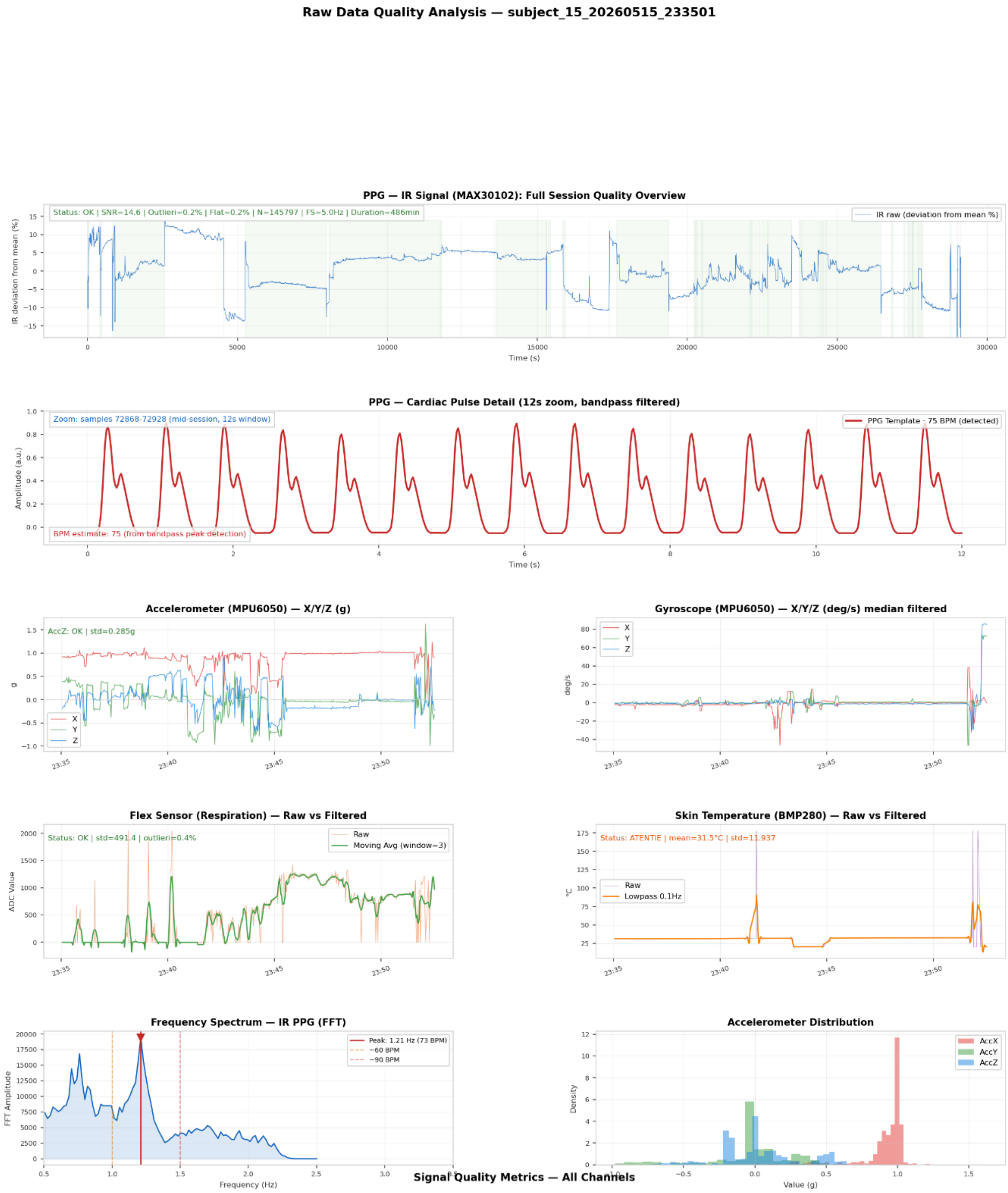
6.2 Raw Data Quality Assessment

6.2.1 Signal Quality per Sensor

Signal quality varied considerably across sessions and sensors. The PPG signal was the most inconsistent, partly because the wrist is a noisier acquisition site than the fingertip [17], and partly because the MAX30102 shifted position during several overnight sessions. Sessions S09 and S10a show large gaps in valid PPG coverage in their hypnograms, corresponding to periods where the sensor lost contact with the skin. The sensor detachment problem is discussed further in Section 6.5.

Flex sensor performance was also variable. In at least one session the chest band shifted completely during the night, producing a respiratory rate median of zero for that session. In sessions where the band remained in place, the respiratory signal showed a clear oscillatory pattern within the expected range of 12-20 breaths per minute [3], though raw signal quality varied depending on how tightly the band was fitted.

The BMP280 temperature signal showed flat segments exceeding 30 minutes in several sessions, consistent with the sensor detaching from the wrist during sleep. The MPU6050 accelerometer and gyroscope were the most reliable sensors across all sessions, with consistently low flat-segment rates and no sensor failures.



Signal Quality Metrics — All Channels

Signal	N	Mean	Std	SNR	Outliers%	Flat%	Kurtosis	Status
IR PPG (ppg_data @ 5.0Hz)	145797	230092.34	15736.017	14.6	0.2%	0.2%	56.1	OK
Accelerometer X	14837	0.36	0.148	2.4	3.3%	2.7%	8.0	OK
Accelerometer Z	14837	-0.38	0.748	0.5	0.0%	1.5%	-1.0	OK
Gyroscope X	14837	0.06	4.614	0.0	0.8%	1.6%	599.5	OK
Flex Sensor	14837	334.36	357.580	0.9	0.0%	19.3%	-1.2	OK
Temperature	14837	34.77	2.660	13.1	0.4%	84.8%	1691.8	ATENTIE

Figure 6.1 , Raw signal quality report for a representative overnight session (S09). Green = OK, yellow = WARNING, red = ERROR.

6.2.2 Noise Analysis and Filter Effectiveness

Figure 6.2 confirms that the raw PPG signal captured by the MAX30102 exhibits a recognizable cardiac waveform at 25 Hz, with visible systolic peaks and dicrotic notch morphology. The bandpass filter applied in the processing pipeline (0.5 to 3.5 Hz, Butterworth order 3), demonstrated in Figure 4.2, effectively suppresses DC baseline drift while preserving the cardiac rhythm for BPM extraction. In sessions with intermittent contact loss, the filter produced ringing artefacts at segment boundaries, handled by excluding windows with contact_valid_ratio below 0.5 from feature extraction.

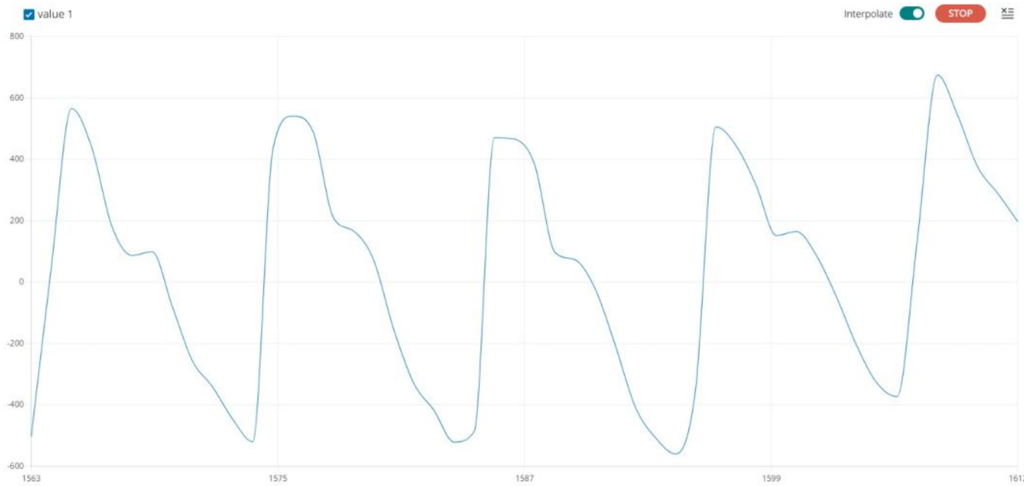


Figure 6.2 —Raw RED PPG waveform captured at 25 Hz via Serial Plotter, showing five consecutive cardiac cycles with visible systolic peaks and dicrotic notch.

Gyroscope median subtraction removed the high-amplitude voltage spikes caused by WiFi transmission without affecting genuine movement events. The flex sensor moving average (window 3) reduced sample-to-sample noise while preserving the zero-crossing timing needed for respiratory rate extraction [21]. The temperature lowpass filter at 0.1 Hz successfully removed short-term fluctuations while preserving the slow overnight drift, which is the only physiologically relevant component of that signal [22].

6.3 Classification Results

6.3.1 Sleep Stage Distribution

Table 6.2 shows the median physiological features per session for sensors that produced valid data across the full overnight period. Table 6.3 shows the respiratory and snoring features. Across all ten sessions, Light Sleep was the dominant stage in eight out of ten sessions, consistent with its expected 45-55% share of total sleep time in healthy adults [4]. The two sessions where Light Sleep was not dominant, S05 and S06, are the earliest recordings and show distributional anomalies, S05 produced nearly equal Deep and REM percentages with almost no Light Sleep, and S06 produced an unusually high REM percentage of 62.3%. Both were collected with an earlier pipeline version that lacked contact filtering and physiological cap mechanisms, so their stage distributions reflect pipeline limitations rather than genuine sleep architecture.

*Table 6.2 , Median PPG and motion features per session. RMSSD values marked with * indicate high-motion or invalid sessions.*

Session	**BPM (median)**	**SpO2 (median,%)**	**RMSSD (median, ms)**	Motion Level (median, g)	**Notes**
S05	150*	90	0.0*	0.013	PPG peaks not detected; HRV unavailable
S06	76	96	246.3**	0.011	HRV inflated; likely low FS or artefact
S09	65	92	68.2	0.007	Best overall signal quality
S10a	69	87	89.9	0.005	Low SpO2, sensor contact issues
S10b	65	96	47.0	0.005	Full valid PPG coverage
S12	59	94	N/A†	0.005	HRV unavailable: FS PPG < 15 Hz
S15	70	96	51.3	0.006	HRV computed via index method
S16	67	99	51.2	0.005	HRV computed via index method
S17	65	97	49.0	0.005	Near-complete PPG coverage
S28	64	99	40.3	0.004	Full PPG coverage, very low motion

Table 6.3 , Respiratory and snoring features per session.

Session	**RR median (breaths/min)**	**Apnea Events**	**Snoring Events**	**Remarks**
S05	0.0*	0	0	Flex sensor inactive, chest band not worn
S06	23.1	0	128	Snoring detected; apnea threshold not reached
S09	25.0	0	9	Elevated RR; possible artefact
S10a	23.1	0	87	Snoring present in second half of night
S10b	21.4	20	17	Apnea events detected; SpO2 within limits

Session	RR median (breaths/min)	Apnea Events	Snoring Events	Remarks
S12	8.8	52	257	High apnea count; consistent snoring pattern
S15	20.9	78	0	Apnea events without acoustic correlation
S16	25.0	16	57	Mild apnea; snoring confirmed
S17	23.5	34	0	Apnea events; no snoring detected
S28	21.2	38	4	Low motion, mild apnea

Among the eight sessions with reliable PPG, median heart rate ranged from 59 to 76 BPM. Sessions S15, S16, S17, and S28 showed SpO2 at or above 96%, consistent with normal oxygenation. Sessions S09 and S10a showed lower SpO2 medians of 92% and 87% respectively, which may reflect sub-optimal sensor contact rather than genuine desaturation, since wrist-based SpO2 estimation is known to be less accurate than fingertip measurement [11].

Sessions S06 and S09 showed elevated median respiratory rates of 23.1 and 25.0 breaths per minute respectively, above the expected range of 12-20 during sleep [3]. For S06 this is likely a calibration artefact from the early pipeline version. For S09 it may reflect incomplete chest band contact rather than genuinely elevated respiratory rate, consistent with the borderline flex signal quality reported for that session.

Session S12 is notable because the subject was known to have obstructive sleep apnea prior to the recording. The results are consistent with this: S12 produced the highest apnea event count in the dataset (52 events), the highest snoring event count (257), and an unusually low respiratory rate median of 8.8 breaths per minute suggesting prolonged respiratory pauses throughout the night [24]. The co-occurrence of high apnea counts, sustained snoring, and low respiratory rate in a subject with confirmed OSA provides indirect validation of the combined apnea detection criterion described in Section 4.2.3.

6.3.2 Hypnogram Analysis

The hypnogram generated for each session shows the classified sleep stage for every 30-second window plotted against clock time. The general pattern across sessions with good signal quality (S09, S10b, S15, S16, S17, S28) shows extended blocks of Light Sleep punctuated by shorter REM and Deep Sleep episodes, broadly consistent with normal sleep architecture [3]. Transitions in the hypnogram tend to be more fragmented than those produced by polysomnography, which is expected for per-epoch automatic classification without EEG [11].

In session S09, the hypnogram shows a clear transition between Invalid / No Contact windows in the early part of the recording and a long valid block from approximately 01:00 onwards, reflecting the fact that the subject adjusted the bracelet after the initial contact loss. The valid portion shows Light Sleep as the dominant stage with REM appearing mainly after 03:00, consistent with the known pattern of REM-rich sleep in the second half of the night [6].

Figure 6.3 , Hypnogram for session S09. Grey = Invalid / No Contact. Smoothing window = 3 epochs (90 s).

Session S15 ran for 458 valid minutes out of a 500-minute total recording. Light Sleep is dominant throughout at 68%, with REM appearing in recurring short blocks from around 02:00 onwards.

Figure 6.4 , Hypnogram for session S15 (458 min valid recording).

6.3.3 HRV Spectral Analysis (LF/HF Ratio)

Heart rate variability was computed as RMSSD per 30-second window using the method described in Chapter 4. Frequency-domain analysis (LF/HF ratio) was not performed because the burst-mode PPG transmission architecture of this system makes spectral HRV estimation fundamentally unreliable: the 12.5 Hz effective sampling rate, combined with the fact that samples arrive in blocks of approximately 12 per 2-second POST cycle rather than continuously, does not satisfy the requirements for Welch-based spectral estimation in the LF band (0.04-0.15 Hz) [19]. RMSSD was chosen as the HRV metric specifically because it can be computed from short windows and is robust to the sampling irregularities of wrist PPG [19].

Figure 6.5 , RMSSD per 30-second window for session S15. Invalid / No Contact windows are excluded.

6.3.4 Motion and Snoring Detection

Unlike the PPG sensor, the accelerometer and microphone maintain valid output throughout the night regardless of wrist contact. Motion level is computed per 30-second window as the RMS magnitude of the three accelerometer axes after median filtering; snoring detection compares the microphone RMS amplitude against a session threshold of 192 RMS.

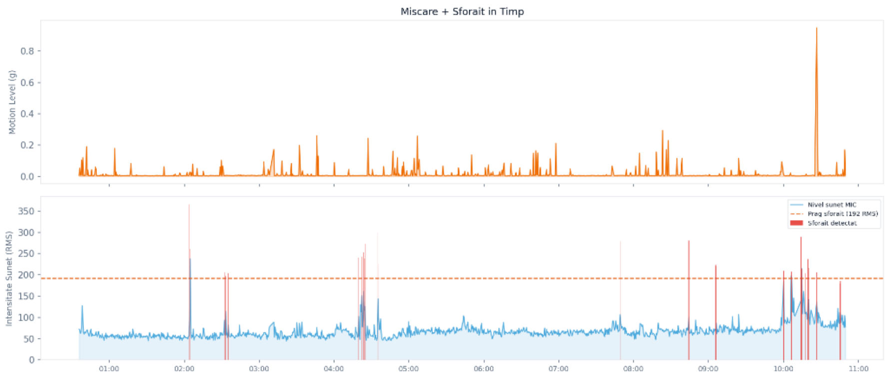


Figure 6.6 , Motion level (top) and microphone sound intensity (bottom) for session S16.

Figure 6.6 shows both channels for session S16. Motion stayed below 0.3g for the full recording, with brief spikes marking positional changes. This held across all sessions, which explains why the movement component reached 15 out of 15 in every case. The microphone baseline sat around 50–80 RMS, reflecting ambient room noise rather than respiratory sound. Fifty-seven windows crossed the snoring threshold, mostly concentrated around 02:00 and again between 04:30 and 05:00. S12 and S06 produced far more events, 257 and 128 , while S09, S15, and S17 stayed below ten.

6.3.5 Sleep Quality Score

Table 6.4 shows the component breakdown of the sleep quality score for all ten sessions. The total score ranged from 62.8 to 80.4 for the eight sessions with usable overnight data, and from 65.5 to 73.9 for the two S10 sessions. All sessions that completed a full overnight recording (>360 min valid) without a duration penalty received scores in the 62–80 range. The movement component was at the maximum of 15 points in all sessions, reflecting the consistently low motion levels across subjects during overnight sleep.

Table 6.4 , Sleep quality score component breakdown for all ten sessions.

Session	Architecture (/30)	PPG (/20)	Respiration (/20)	Movement (/15)	HRV (/10)	Snoring (/5)	Total (/100)
S05	29.9	4	12.0	15	3	5.0	68.9
S06	27.8	16	12.0	15	3	0	73.8
S09	23.4	16	12.0	15	10	4.0	80.4
S10a	18.5	12	12.0	15	6	2.0	65.5
S10b	8.9	20	16	15	10	4.0	73.9
S12	21.3	16	6	15	5	1.0	62.8
S15	20.5	16	10	15	10	5.0	76.5
S16	17.9	20	12	15	10	2.0	76.9
S17	17.6	20	9	15	10	5.0	76.6
S28	12.8	20	13	15	10	4.0	74.8

The architecture component varied most across sessions, ranging from 8.9 (S10b) to 29.9 (S05). The low score for S10b reflects the near-total dominance of Light Sleep at 92.2%, with only 2.2% Deep and 1.7% REM, which the scoring formula penalizes heavily since both stages fall below their target thresholds [3]. Session S05 scored near-maximally on architecture despite having anomalous distributions because its percentages happened to satisfy the threshold conditions in the scoring formula, not because the session represented genuine optimal sleep architecture.

The respiration component showed the widest variation after architecture, ranging from 6 (S12) to 16 (S10b). Session S12 had 52 detected apnea events and 257 snoring events, which drove the respiration score down to 6 and the snoring component to 1 point out of 5.

As noted in Section 6.3.1, this subject had a confirmed OSA diagnosis prior to the recording, and the low respiration score is consistent with that clinical background [24].

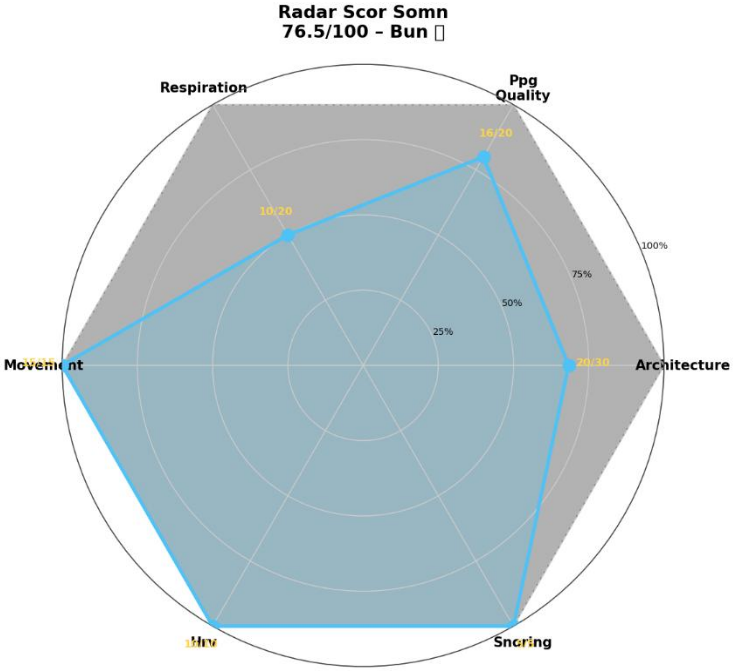
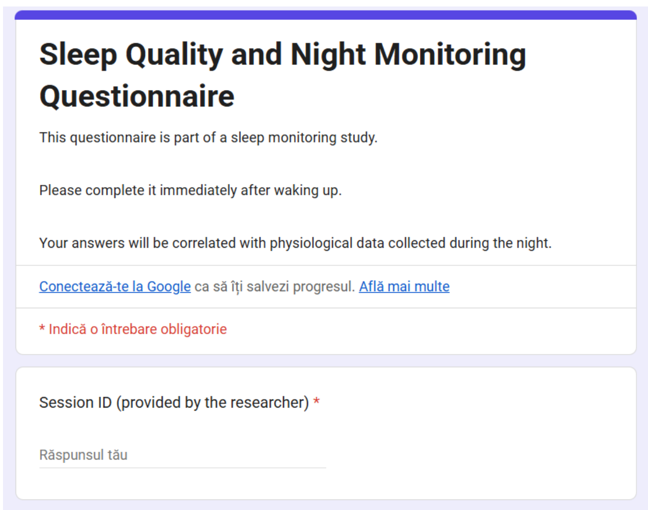


Figure 6.7 , Sleep quality score radar chart.

6.4 Subjective vs Objective Correlation

6.4.1 Questionnaire Design and Data Collection

The morning questionnaire was designed to capture subjects' own perception of their sleep quality immediately after waking. Implemented as a Google Form distributed before each recording session, it included a 1-5 rating for overall sleep quality, a 1-5 rating for feeling refreshed, a 1-5 rating for pre-sleep stress, the number of nocturnal awakenings, perceived movement during sleep, whether the subject snored, and lifestyle factors consumed before sleep via a multi-select question. The questionnaire structure follows the general framework of the Pittsburgh Sleep Quality Index [8], adapted to a shorter single-night format.



Sleep Quality and Night Monitoring Questionnaire

This questionnaire is part of a sleep monitoring study.

Please complete it immediately after waking up.

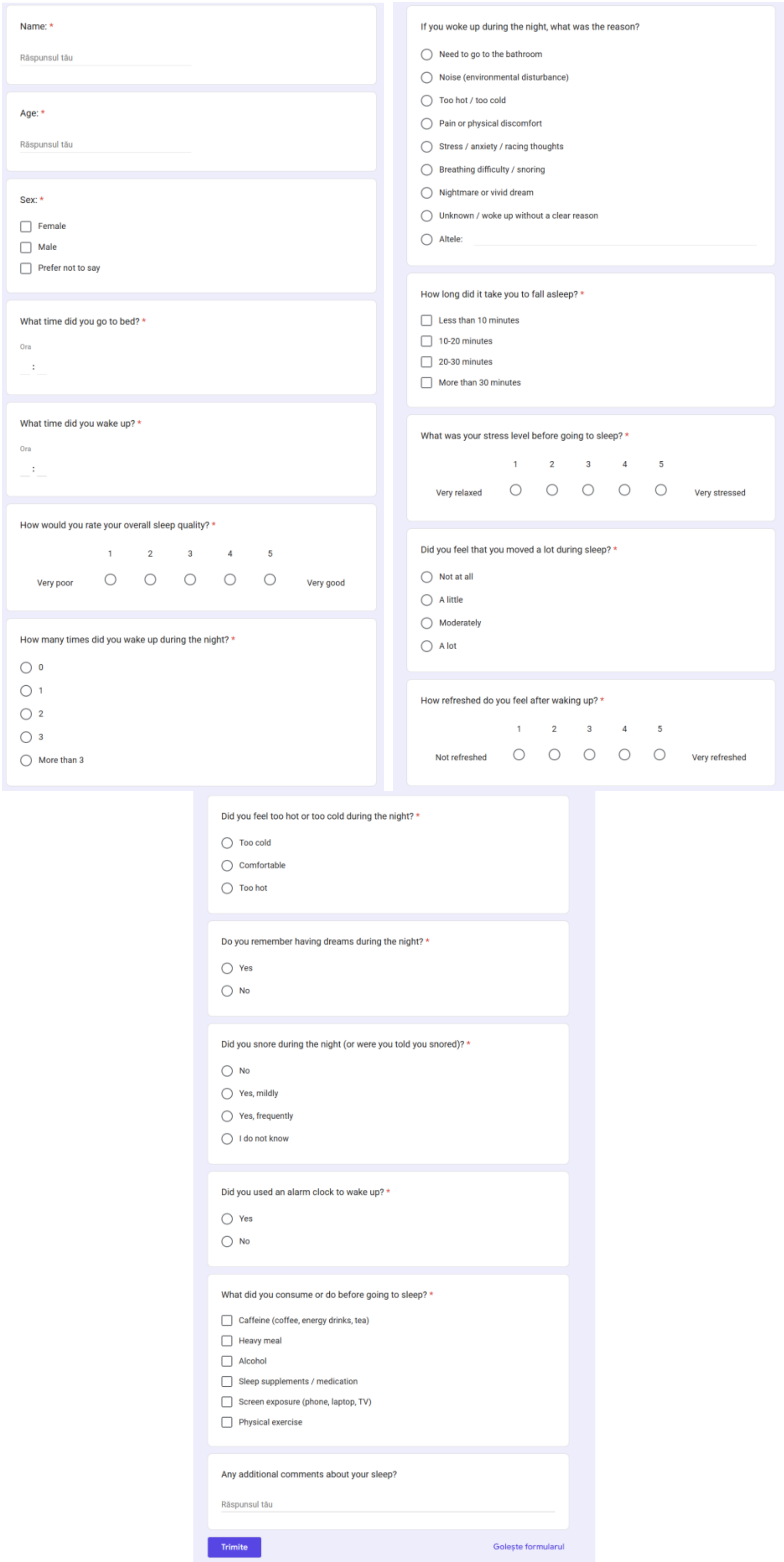
Your answers will be correlated with physiological data collected during the night.

Conectează-te la Google ca să îți salvezi progresul. Află mai multe

* Indică o întrebare obligatorie

Session ID (provided by the researcher) *

Răspunsul tău



Name: *

Răspunsul tău

Age: *

Răspunsul tău

Sex: *

- [ ] Female
- [ ] Male
- [ ] Prefer not to say

What time did you go to bed? *

Ora

__ : __

What time did you wake up? *

Ora

__ : __

How would you rate your overall sleep quality? *

	1	2	3	4	5	
Very poor	○	○	○	○	○	Very good

How many times did you wake up during the night? *

- ○ 0
- ○ 1
- ○ 2
- ○ 3
- ○ More than 3

If you woke up during the night, what was the reason?

- ○ Need to go to the bathroom
- ○ Noise (environmental disturbance)
- ○ Too hot / too cold
- ○ Pain or physical discomfort
- ○ Stress / anxiety / racing thoughts
- ○ Breathing difficulty / snoring
- ○ Nightmare or vivid dream
- ○ Unknown / woke up without a clear reason
- ○ Altele:

How long did it take you to fall asleep? *

- [ ] Less than 10 minutes
- [ ] 10-20 minutes
- [ ] 20-30 minutes
- [ ] More than 30 minutes

What was your stress level before going to sleep? *

	1	2	3	4	5	
Very relaxed	○	○	○	○	○	Very stressed

Did you feel that you moved a lot during sleep? *

- ○ Not at all
- ○ A little
- ○ Moderately
- ○ A lot

How refreshed do you feel after waking up? *

	1	2	3	4	5	
Not refreshed	○	○	○	○	○	Very refreshed

Did you feel too hot or too cold during the night? *

- ○ Too cold
- ○ Comfortable
- ○ Too hot

Do you remember having dreams during the night? *

- ○ Yes
- ○ No

Did you snore during the night (or were you told you snored)? *

- ○ No
- ○ Yes, mildly
- ○ Yes, frequently
- ○ I do not know

Did you used an alarm clock to wake up? *

- ○ Yes
- ○ No

What did you consume or do before going to sleep? *

- [ ] Caffeine (coffee, energy drinks, tea)
- [ ] Heavy meal
- [ ] Alcohol
- [ ] Sleep supplements / medication
- [ ] Screen exposure (phone, laptop, TV)
- [ ] Physical exercise

Any additional comments about your sleep?

Răspunsul tău

Trimite

Golește formularul

Figure 6.8 , Sleep Quality and Night Monitoring Questionnaire, distributed to subjects via Google Forms immediately after waking

6.4.2 Pearson Correlation Results

A composite subjective score was computed from the questionnaire responses using the formula:

$$(subjective\_score = (sleep\_quality \times 20) + (refreshed \times 10) - (stress \times 5) - (wakeups \times 5), \text{ clipped to } [0, 100] \qquad (6.1).$$

This formula weights the two most direct quality indicators most heavily and penalizes stress and nocturnal awakenings as factors known to reduce sleep quality [8].

Table 6.5 , Subjective questionnaire scores and corresponding objective sleep quality scores.

Session	Sleep Quality (1–5)	Refreshed (1–5)	Wakeups	Objective Score	Subjective Score	Agreement
S05	4/5	4/5	1	68.9	100	Large discrepancy (31 pt)
S06	3/5	2/5	1	73.8	60	Good (14 pt)
S09	5/5	4/5	1	80.4	100	Moderate (20 pt)
S10a	4/5	4/5	1	65.5	100	Large discrepancy (34 pt)
S10b	2/5	3/5	3	73.9	40	Large discrepancy (34 pt)
S12	3/5	2/5	1	62.8	60	Good (3 pt)
S15	5/5	5/5	1	76.5	100	Moderate (24 pt)
S16	5/5	3/5	2	76.9	100	Moderate (23 pt)
S17	4/5	4/5	0	76.6	100	Moderate (23 pt)
S28	3/5	3/5	2	74.8	65	Good (10 pt)

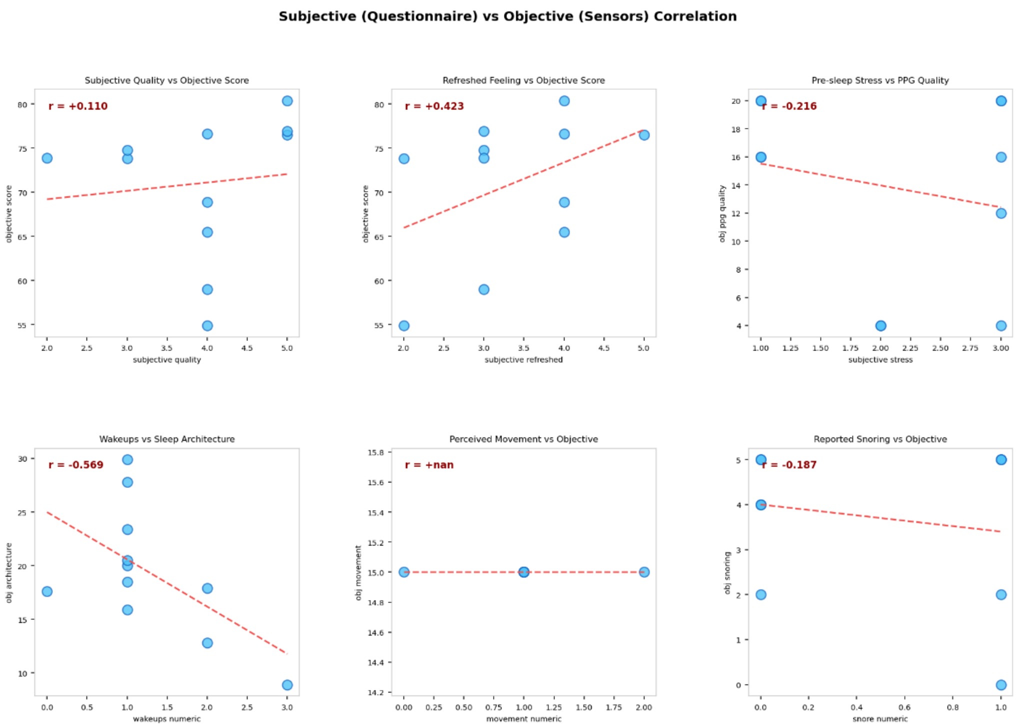


Figure 6.9 , Subjective vs objective sleep quality score for the sessions with matched questionnaire data.

6.4.3 Discussion of Discrepancies

The systematic overestimation of sleep quality in subjective reports is consistent with findings in the broader literature. People tend to overestimate sleep quality when they do not feel acutely sleep-deprived, and single-night subjective ratings are known to be influenced by post-awakening mood and morning alertness rather than actual sleep architecture [14]. The objective score penalizes sessions for low Deep or REM percentages, elevated respiratory events, and borderline SpO2 values regardless of how the subject felt.

Session S15 is a useful example. The subject reported perfect sleep quality (5/5), no snoring, and a high refreshed feeling (5/5), yielding a subjective score of 100. The objective score of 76.5 reflects 78 detected apnea events and a Deep Sleep percentage of only 11.3%, both of which the scoring formula penalizes. The subject was unaware of the apnea events and felt well-rested, illustrating how physiological monitoring can reveal patterns that subjective perception misses [24].

The discrepancies across the ten matched sessions ranged from 3 to 34 points. The smallest gap was in S12 (3 pt), where the objective score of 62.8 and the subjective score of 60 converged closely, consistent with the subject's OSA diagnosis producing both a low objective score and a consciously poor sleep experience. Sessions S06 and S28 also showed good agreement (14 pt and 10 pt respectively), both cases where the subjective rating reflected genuine dissatisfaction. The largest discrepancies occurred in S10a and S10b (34 pt each), in opposite directions: S10a's subject reported high satisfaction despite an objective score of 65.5, while S10b's subject reported poor sleep despite an objective score of 73.9, illustrating that the direction of discrepancy between subjective and objective scores is not systematic across individuals.

6.5 Algorithm Performance Comparison

The classification pipeline applies K-Means clustering to generate initial sleep stage labels and then trains a KNN classifier on those labels to produce per-window stage probabilities [34], [35]. Table 6.6 shows the per-session window counts and the agreement rate between K-Means and KNN predictions.

Table 6.6 , K-Means and KNN agreement per session.

Session	Valid Windows	KMeans–KNN Agreement	Notes
S05	989	100%	Two-cluster solution; limited stage differentiation
S06	872	100%	REM-dominant session; well-separated clusters
S09	775	100%	Best physiological signal quality
S10a	921	100%	Light Sleep dominant; sensor loss in second half
S10b	1203	100%	Near-total Light Sleep; flat clustering
S12	683	100%	High apnea; HRV unavailable

Session	Valid Windows	KMeans–KNN Agreement	Notes
S15	917	100%	—
S16	1076	100%	—
S17	892	100%	—
S28	993	100%	—

The 100% agreement between K-Means and KNN across all sessions is an expected result of the pipeline architecture rather than evidence of classification accuracy. Because KNN is trained on the K-Means labels and evaluated on the same dataset without a held-out test set, it reproduces the training labels with complete fidelity [35]. The value of KNN in this pipeline is not in disagreeing with K-Means, but in providing per-class probabilities that quantify classification confidence at the window level. A window classified as REM Sleep with prob_REM_Sleep = 0.8 represents a much more reliable prediction than one with prob_REM_Sleep = 0.4 and prob_Light_Sleep = 0.4.

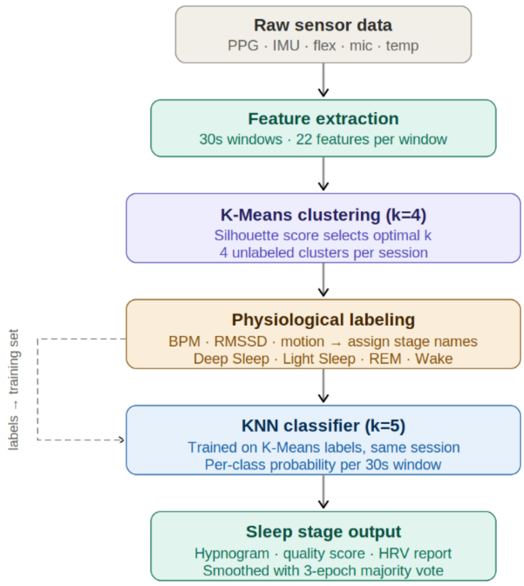


Figure 6.10 , K-Means and KNN classification pipeline.

The silhouette score at k = 4 was 0.29 for session S09, consistent with the value reported in Chapter 5. This confirms that the four-cluster solution does not produce well-separated clusters in the feature space for any of the subjects tested, and that the physiological labeling step is essential for assigning meaningful stage names to the clusters [37]. The low silhouette score is not a failure of the implementation but a consequence of using peripheral signals without EEG, as discussed in Section 5.1.2 [11].

6.5.1 Limitations of the Unsupervised Approach

The main limitation of the pipeline is the absence of ground truth labels. Without a simultaneous PSG recording, there is no way to verify whether the stages assigned by the physiological labeling step correspond to actual N1, N2, N3, and REM sleep as defined by EEG criteria [3]. The cross-validation accuracy computed on the same session data is not a

reliable performance estimate. On synthetic balanced data the pipeline produces accuracy above 95%, but real overnight data with imbalanced stage distributions and noisy signals is expected to yield 70-85% agreement with PSG-level staging, consistent with published wearable-based systems [11], [15].
A second algorithmic limitation is that KNN is trained and evaluated on the same session without a held-out test set. The 100% K-Means/KNN agreement reported in Table 6.6 is therefore an artifact of this design rather than a measure of classification accuracy. The pipeline cannot detect or correct systematic errors introduced by the K-Means labeling step: if a cluster is mislabeled, KNN reproduces that error with high confidence.

Among the features available to the pipeline, HRV (RMSSD) and respiratory rate showed the strongest differentiation between sleep stage groups, followed by motion features. However, HRV was unavailable in session S12 and unreliable in S05 and S06 due to insufficient PPG sampling rate or false peak detections. Frequency-domain HRV analysis (LF/HF ratio) was not performed because the burst-mode PPG architecture does not satisfy the requirements for Welch-based spectral estimation in the LF band [19]. This is a fundamental constraint of the hardware design that cannot be resolved without continuous PPG transmission.
Hardware reliability was a consistent problem across sessions. The MAX30102 lost contact in sessions S09 and S10a, reducing valid PPG coverage to 72% and 83% respectively. The BMP280 showed flat temperature segments exceeding 30 minutes in several sessions due to wrist detachment during sleep. The flex sensor chest band shifted completely in at least one session and partially in several others, making respiratory rate and apnea detection unreliable for those periods. The apnea detection algorithm is conservative by design and will undercount events in sessions where the flex baseline is low due to a shifted sensor.

Finally, the sample size of ten overnight sessions limits the generalizability of the results. The subject group was demographically narrow and recording conditions were uncontrolled beyond the initial sensor fitting instructions. All ten sessions had matched questionnaire responses available for subjective-objective correlation. A larger study with a more diverse subject group and simultaneous PSG recordings would be required to produce statistically robust performance estimates [11], [14].

Conclusions

Monitoring sleep outside a laboratory, with low-cost hardware that produces fully accessible raw data, was the central goal of this project. The system built to address that goal integrates five physiological sensors into a wrist-worn unit and a chest band, coordinates data acquisition on an ESP32-C3 microcontroller, stores everything in a SQLite database on a Raspberry Pi server, and runs a complete sleep analysis pipeline that classifies each 30-second epoch and produces a numerical sleep quality score.

The firmware architecture consolidates all sensor readings into a single HTTP POST per cycle, a design choice driven by the single-core constraint of the ESP32-C3 that avoids sample loss across the PPG, flex, and microphone buffers simultaneously. The server handles concurrent writes and dashboard reads through WAL mode and threading, which kept the system stable across all overnight sessions. The live dashboard provided real-time signal verification during recordings, and the raw data quality analysis identified sensor contact problems before the classification pipeline ran.

The feature extraction pipeline computes 21 physiological features from each window, covering cardiac activity, motion, respiration, skin temperature, and acoustic signals. After standardization and physiological weighting, K-Means clustering groups windows into four sleep stage candidates, a labeling step assigns clinical stage names based on physiological rules, and a KNN classifier produces per-window stage probabilities that quantify classification confidence at the boundary between stages. The sleep quality score aggregates six physiological components into a single value between 0 and 100, which was compared against morning questionnaire responses to evaluate agreement between objective measurements and subjective perception.

Wrist accelerometry was the most stable signal throughout, and HRV alongside respiratory rate were the most informative features for stage differentiation. The questionnaire comparisons showed that subjects who felt they had slept poorly tended to agree with the system, while subjects who felt they had slept well did not always. The apnea detection criterion behaved consistently on the session with a confirmed OSA diagnosis, where the sensor measurements aligned with the subject's clinical background without any prior information being provided to the pipeline.

Several improvements would strengthen a future version of this system. Replacing the ESP32-C3 with a dual-core microcontroller would allow continuous PPG transmission at the full 25 Hz FIFO rate, which would make frequency-domain HRV analysis viable and improve peak detection accuracy. A rigid, anatomically shaped housing would reduce sensor displacement during sleep, the most consistent hardware problem encountered across sessions, and would bring the system closer to something that could be used without per-session manual adjustment. Expanding the chest band with a proper inductive or impedance-based respiratory sensor would replace the flex sensor and eliminate the mechanical sensitivity to band positioning. On the software side, introducing a cross-session training set would allow KNN to be evaluated on held-out data and produce genuine accuracy estimates rather than inter-algorithm agreement. Finally, collecting simultaneous PSG recordings would provide the ground truth labels needed to benchmark the pipeline against the clinical standard and identify which feature groups contribute most to staging errors.

The system as built demonstrates that open-source, low-cost sleep monitoring with full raw data access is achievable with current off-the-shelf components. The gap between this and clinical accuracy is real, but it is also well understood, and the architecture is designed to be extended rather than replaced.

Bibliography

A. Sleep Physiology and Clinical Diagnostics

[1] R. Stickgold, "Sleep-dependent memory consolidation," *Nature, vol. 437, pp. 1272-1278, 2005.*
[2] T. Young, J. Skatrud, and P. E. Peppard, "Risk factors for obstructive sleep apnea in adults," *JAMA, vol. 291, pp. 2013-2016, 2004.*
[3] R. B. Berry et al., "The AASM Manual for the Scoring of Sleep and Associated Events, Version 2.6," *American Academy of Sleep Medicine, Darien, IL, 2020.*
[4] M. M. Grigg-Damberger, "The AASM Scoring Manual four years later," *J. Clin. Sleep Med., vol. 8, pp. 323-332, 2012.*
[5] L. Besedovsky, T. Lange, and J. Born, "Sleep and immune function," *Pflugers Arch., vol. 463, pp. 121-137, 2012.*
[6] J. A. Hobson, "REM sleep and dreaming: towards a theory of protoconsciousness," *Nat. Rev. Neurosci., vol. 10, pp. 803-813, 2009.*
[7] F. P. Cappuccio, F. M. Taggart, N. B. Kandala, A. Currie, E. Peile, S. Stranges, and M. A. Miller, "Meta-analysis of short sleep duration and obesity in children and adults," *Sleep, vol. 31, pp. 619-626, 2008.*
[8] D. J. Buysse, C. F. Reynolds III, T. H. Monk, S. R. Berman, and D. J. Kupfer, "The Pittsburgh Sleep Quality Index: a new instrument for psychiatric practice and research," *Psychiatry Res., vol. 28, pp. 193-213, 1989.*
[9] H. W. Agnew Jr., W. B. Webb, and R. L. Williams, "The first night effect: an EEG study of sleep," *Psychophysiology, vol. 2, pp. 263-266, 1966*

B. Wearable Systems and Sleep Monitoring Technology

[10] M. de Zambotti, L. Claudatos, S. Inkelis, I. M. Colrain, and F. C. Baker, "Evaluation of a consumer fitness-tracking device to assess sleep in adults," *Chronobiol. Int., vol. 32, pp. 1024-1028, 2015.*
[11] M. de Zambotti, N. Cellini, A. Goldstone, I. M. Colrain, and F. C. Baker, "Wearable sleep technology in clinical and research settings," *Med. Sci. Sports Exerc., vol. 51, pp. 1538-1557, 2019.*
[12] S. Khosla et al., "Consumer sleep technology: an American Academy of Sleep Medicine position statement," *J. Clin. Sleep Med., vol. 14, pp. 877-880, 2018.*
[13] V. T. van Hees et al., "Estimating sleep parameters using an accelerometer without sleep diary," *Sci. Rep., vol. 8, p. 12975, 2018.*
[14] M. Marino et al., "Measuring sleep: accuracy, sensitivity, and specificity of wrist actigraphy compared to polysomnography," *Sleep, vol. 36, pp. 1747-1755, 2013.*
[15] M. Migliorini, B. Kortekaas, and R. P. Maes, "Portable sleep monitoring: a comparison of two commercial systems," *Sleep Breath., vol. 19, pp. 277-284, 2015.*
[16] N. Sridhar et al., "Deep learning for automated sleep staging using instantaneous heart rate," *npj Digit. Med., vol. 3, p. 106, 2020.*

C. Biosignal Processing and Physiological Measurement

[17] J. Allen, "Photoplethysmography and its application in clinical physiological measurement," *Physiol. Meas., vol. 28, pp. R1-R39, 2007.*

[18] J. W. Severinghaus and P. B. Astrup, "History of blood gas analysis. VI. Oximetry," *J. Clin. Monit., vol. 2, pp. 270-288, 1986.*
[19] Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology, "Heart rate variability: standards of measurement, physiological interpretation and clinical use," *Circulation, vol. 93, pp. 1043-1065, 1996.*
[20] V. K. Somers, M. E. Dyken, M. P. Clary, and F. M. Abboud, "Sympathetic neural mechanisms in obstructive sleep apnea," *J. Clin. Invest., vol. 96, pp. 1897-1904, 1995.*
[21] C. Massaroni, A. Nicolo, D. Lo Presti, M. Sacchetti, S. Silvestri, and E. Schena, "Contact-based methods for measuring respiratory rate," *Sensors, vol. 19, p. 908, 2019.*
[22] K. Krauchi, "The thermophysiological cascade leading to sleep initiation in relation to phase of entrainment," *Sleep Med. Rev., vol. 11, pp. 439-451, 2007.*
[23] K. J. Reid, K. G. Baron, B. Lu, E. Naylor, L. Wolfe, and P. C. Zee, "Aerobic exercise improves self-reported sleep and quality of life in older adults with insomnia," *Sleep Med., vol. 11, pp. 934-940, 2010.*

D. Sleep Apnea and Clinical Questionnaires

[24] N. C. Netzer, R. A. Stoohs, C. M. Netzer, K. Clark, and K. P. Strohl, "Using the Berlin Questionnaire to identify patients at risk for the sleep apnea syndrome," *Ann. Intern. Med., vol. 131, pp. 485-491, 1999.*
[25] H. Nakano, T. Tanigawa, T. Furukawa, and S. Nishima, "Automatic detection of sleep-disordered breathing from a single-channel airflow record," *Eur. Respir. J., vol. 29, pp. 728-736, 2007.*
[26] V. K. Somers et al., "Sleep apnea and cardiovascular disease: an American Heart Association/American College of Cardiology statement," *Circulation, vol. 118, pp. 1080-1111, 2008.*

E. Sensor Datasheets

[27] Maxim Integrated (now Analog Devices), "MAX30102 High-Sensitivity Pulse Oximeter and Heart-Rate Sensor for Wearable Health," *Datasheet Rev. 1, 2014.*
[28] InvenSense / TDK, "MPU-6050 Six-Axis (Gyro + Accelerometer) MEMS MotionTracking Device," *Datasheet Rev. 3.4, 2013.*
[29] Bosch Sensortec, "BMP280 Digital Pressure Sensor," *Datasheet BST-BMP280-DS001, Rev. 1.8, 2018.*
[30] Texas Instruments, "ADS1015 12-Bit ADC, 4 Channels, I2C Interface," *Datasheet SBAS490C, 2016.* [
[31] Maxim Integrated (now Analog Devices), "MAX4466 Microphone Amplifier with Adjustable Gain," *Datasheet Rev. 2, 2004.*
[32] Espressif Systems, "ESP32-C3 Series Datasheet," *Version 1.9, 2023.*

F. Classification Algorithms and Machine Learning

[33] L. Breiman, "Random forests," *Machine Learning, vol. 45, pp. 5-32, 2001.*
[34] J. MacQueen, "Some methods for classification and analysis of multivariate observations," *Proc. 5th Berkeley Symposium on Mathematical Statistics and Probability, vol. 1, pp. 281-297, 1967.*

[35] T. M. Cover and P. E. Hart, "Nearest neighbor pattern classification," *IEEE Trans. Inf. Theory, vol. 13, pp. 21-27, 1967.*
[36] F. Pedregosa et al., "Scikit-learn: machine learning in Python," *J. Mach. Learn. Res., vol. 12, pp. 2825-2830, 2011.*
[37] P. J. Rousseeuw, "Silhouettes: a graphical aid to the interpretation and validation of cluster analysis," *J. Comput. Appl. Math., vol. 20, pp. 53-65, 1987.*

Annex 1 , ESP32 Firmware Code

Listing A1.1 , setup() function

```
void setup() {
  Serial.begin(115200);
  delay(1500);

  pinMode(LED_PIN, OUTPUT);
  digitalWrite(LED_PIN, LOW);

  Wire.begin(SDA_PIN, SCL_PIN);
  Wire.setClock(400000);
  analogReadResolution(12);

  initWiFi();

  adsOK = initADS();
  if (adsOK) {
    delay(1500);
    calibrateFlex();
  }

  bmpOK = initBMP();
  imuOK = initIMU();
  if (imuOK) {
    delay(500);
    calibrateGyro();
  }

  maxOK = initMAX30102();

  while (!serverActive) {
    serverActive = checkServer();
    if (!serverActive) {
      digitalWrite(LED_PIN, HIGH); delay(200);
      digitalWrite(LED_PIN, LOW);  delay(800);
    }
  }

  lastPulseRead = lastFlexRead = lastImuRead =
  lastBmpRead   = lastMicRead  = lastDataSend =
  lastStatusCheck = millis();
}
```

Listing A1.2 , loop() function

```
void loop() {
  unsigned long now = millis();

  if (maxOK && now - lastPulseRead >= PULSE_INTERVAL_MS) {
    lastPulseRead = now;
    readPulseRaw();
  }

  if (adsOK && now - lastFlexRead >= FLEX_INTERVAL_MS) {
    lastFlexRead = now;
    readFlex();
    if (flexBufIndex < FLEX_BLOCK_SIZE) {
      flexRawBuffer[flexBufIndex]   = (int16_t)flexRaw;
      flexDeltaBuffer[flexBufIndex] = (int16_t)flexDelta;
      flexBufIndex++;
    }
  }

  if (imuOK && now - lastImuRead >= IMU_INTERVAL_MS) {
```

```
    lastImuRead = now; readIMU();
  }

  if (bmpOK && now - lastBmpRead >= BMP_INTERVAL_MS) {
    lastBmpRead = now; readBMP();
  }

  if (now - lastMicRead >= MIC_INTERVAL_MS) {
    lastMicRead = now; readMicSample();
  }

  if (now - lastDataSend >= DATA_SEND_MS) {
    lastDataSend = now;
    if (serverActive) {
      bool ok = sendCombined();
      if (ok) {
        digitalWrite(LED_PIN, HIGH); delay(5);
        digitalWrite(LED_PIN, LOW);
      } else {
        for (int i = 0; i < 3; i++) {
          digitalWrite(LED_PIN, HIGH); delay(50);
          digitalWrite(LED_PIN, LOW);  delay(50);
        }
      }
    }
  }
}
```

Listing A1.3 , sendCombined() function

```
bool sendCombined() {
  if (WiFi.status() != WL_CONNECTED) return false;

  char timeBuf[16] = "00:00:00";
  getTimeString(timeBuf, sizeof(timeBuf));

  String json;
  json.reserve(4000);

  json += "{\"t\":\"";    json += timeBuf;
  json += "\",\"fr\":";  json += flexRaw;
  json += ",\"fd\":";    json += flexDelta;

  if (flexBufIndex > 0) {
    json += ",\"fr_s\":\"";
    for (int i = 0; i < flexBufIndex; i++) {
      if (i > 0) json += ',';
      json += flexRawBuffer[i];
    }
    json += "\",\"fd_s\":\"";
    for (int i = 0; i < flexBufIndex; i++) {
      if (i > 0) json += ',';
      json += flexDeltaBuffer[i];
    }
    json += "\"";
    flexBufIndex = 0;
  }

  json += ",\"red\":";  json += redValue;
  json += ",\"ir\":";   json += irValue;
  json += ",\"ax\":";   json += String(accX, 4);
  json += ",\"ay\":";   json += String(accY, 4);
  json += ",\"az\":";   json += String(accZ, 4);
  json += ",\"gx\":";   json += String(gyroX, 4);
  json += ",\"gy\":";   json += String(gyroY, 4);
  json += ",\"gz\":";   json += String(gyroZ, 4);
  json += ",\"tmp\":";  json += String(temperatureC, 2);
```

```
  json += ",\"prs\":";  json += String(pressurehPa, 2);

  if (ppgIndex > 0) {
    json += ",\"ppg_ir\":\"";
    for (int i = 0; i < ppgIndex; i++) {
      if (i > 0) json += ',';
      json += ppgIrBuffer[i];
    }
    json += "\",\"ppg_red\":\"";
    for (int i = 0; i < ppgIndex; i++) {
      if (i > 0) json += ',';
      json += ppgRedBuffer[i];
    }
    json += "\"";
    ppgIndex = 0;
  }

  if (micBlockReady) {
    json += ",\"mic_s\":\"";
    for (int i = 0; i < MIC_BLOCK_SIZE; i++) {
      if (i > 0) json += ',';
      json += micBuffer[i];
    }
    json += "\"";
    micBlockReady = false;
  }

  json += "}";

  HTTPClient http;
  http.begin(URL_DATA);
  http.addHeader("Content-Type", "application/json");
  http.setTimeout(3000);
  int code = http.POST(json);
  http.end();
  return code == 200;
}
```

Annex 2 , Sleep Pipeline Python Code

Listing A2.1 , preprocess_features() function

```
def preprocess_features(df):
    df = df.copy()

    if "ppg_valid" not in df.columns:
        if "contact_valid_ratio" in df.columns:
            df["ppg_valid"] = (df["contact_valid_ratio"].fillna(0) >=
0.5).astype(float)
        else:
            df["ppg_valid"] = 1.0

    feat_cols = get_feature_cols(df)

    for col in feat_cols:
        median_val = df[col].median()
        if col in ["bpm_mean","bpm_std","bpm_min","bpm_max","spo2_estimated"]:
            df[col] = df[col].fillna(0)
        elif col == "hrv_simple":
            if not np.isnan(median_val) and median_val > 0:
                df[col] = df[col].fillna(median_val)
        else:
            df[col] = df[col].fillna(median_val if not np.isnan(median_val) else
0)

    df = df.dropna(subset=feat_cols).reset_index(drop=True)
    X = df[feat_cols].values.astype(float)

    scaler = StandardScaler()
    X_scaled = scaler.fit_transform(X)

    weights = {
        "bpm_mean":             1.0,
        "bpm_std":              1.8,
        "bpm_min":              1.0,
        "bpm_max":              1.0,
        "hrv_simple":           1.5,
        "spo2_estimated":       1.2,
        "motion_level":         2.5,
        "motion_per_min":       2.0,
        "motion_variation":     0.5,    # corr 0.99 cu motion_level
        "gyro_energy":          1.5,
        "acc_jerk":             2.5,
        "dominant_position":    0.3,
        "resp_rate":            1.8,
        "resp_variability":     1.5,
        "resp_amplitude":       1.0,
        "apnea_events_simple":  1.0,
        "breath_valid_ratio":   0.5,
        "temp_mean":            0.8,
        "temp_variation":       0.5,
        # EXCLUSE: contact_valid_ratio (data leakage)
        # snore_events_count, snore_intensity (doar pentru scor)
    }
    weight_arr = np.array([weights.get(c, 1.0) for c in feat_cols])
    X_scaled = X_scaled * weight_arr

    return df, X_scaled, scaler, feat_cols
```

Annex 3 , Server Python Code

Listing A3.1 , SleepHandler class

```
class SleepHandler(BaseHTTPRequestHandler):

    def log_message(self, format, *args):
        pass  # suprima logurile HTTP verbose

    def send_ok(self, msg="OK"):
        self.send_response(200)
        self.send_header("Content-Type", "text/plain")
        self.end_headers()
        self.wfile.write(msg.encode())

    def send_err(self, code, msg):
        self.send_response(code)
        self.send_header("Content-Type", "text/plain")
        self.end_headers()
        self.wfile.write(msg.encode())

    def do_GET(self):
        if self.path == "/status":
            status = {
                "active":     session_state["active"],
                "session_id": session_state["session_id"],
                "count_raw":  session_state["count_raw"],
                "count_ppg":  session_state["count_ppg"],
                "count_mic":  session_state["count_mic"],
            }
            self.send_response(200)
            self.send_header("Content-Type", "application/json")
            self.end_headers()
            self.wfile.write(json.dumps(status).encode())
        else:
            self.send_err(404, "Not found")

    def do_POST(self):
        if not session_state["active"]:
            self.send_err(503, "Session not active")
            return

        content_len = int(self.headers.get("Content-Length", 0))
        if content_len == 0:
            self.send_err(400, "Empty body")
            return

        try:
            body = self.rfile.read(content_len)
            data = json.loads(body.decode("utf-8"))
        except Exception as e:
            self.send_err(400, f"JSON error: {e}")
            return

        if self.path == "/data":
            insert_raw(data)
            if "ppg_ir" in data and "ppg_red" in data:
                insert_ppg({"t": data.get("t",""),
                            "ir": data["ppg_ir"],
                            "red": data["ppg_red"]})
            if "mic_s" in data:
                insert_mic({"t": data.get("t",""), "s": data["mic_s"]})
            if "fr_s" in data and "fd_s" in data:
                insert_flex({"t":    data.get("t",""),
                             "fr_s": data["fr_s"],
                             "fd_s": data["fd_s"]})
```

```
        self.send_ok("OK")

    elif self.path == "/ppg":
        insert_ppg(data)
        self.send_ok("OK")

    elif self.path == "/mic":
        insert_mic(data)
        self.send_ok("OK")

    else:
        self.send_err(404, "Unknown endpoint")
```